



Chalcopyrite, sphalerite, and pyrite chemistry in stratiform sediment-hosted Cu(-Co) metallogenic districts: trace element characteristics and factors controlling polymetallic mineralization

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Abstract

Stratiform sediment-hosted Cu deposits are significant global sources of Cu and other important metals. The Polish Kupferschiefer produces Ag, Au, Pb, Ni, Se, and Re as by-products, whereas Co is one of the most important metals in the stratiform sediment-hosted Cu-Co deposits of the Central African Copperbelt and the Namibian Dolostone Ore Formation deposit. This study combines new and published laser ablation inductively coupled plasma mass spectrometry sulfide trace element data from these stratiform sediment-hosted copper districts. All the investigated districts exhibit sulfides occurring as disseminations and within later veins. Chalcopyrite, sphalerite, and pyrite trace element contents vary significantly between the metallogenic districts as well as between different ore stages. Random Forest discriminates the stratiform sediment-hosted Cu(-Co) districts based on trace element geochemistry. High Ag and Tl in chalcopyrite is attributed to the Polish Kupferschiefer, Ga and Ge to the Katanga Copperbelt, and Zn and In to the Dolostone Ore Formation deposit. Sphalerite from the Polish Kupferschiefer and the Dolostone Ore Formation deposit can be distinguished on the basis of the Fe and Cd contents. Cobalt and As are significantly elevated in pyrite from the Katanga Copperbelt and Mn in pyrite from the Dolostone Ore Formation deposit. The trace element contents also show that the stratiform sediment-hosted Cu(-Co) deposit sulfide data cluster separately from other deposit types. The variation in sulfide trace element contents between the three investigated stratiform sediment-hosted Cu(-Co) districts suggests that sulfide chemistry is related to the geology of the host basin and the nature of the underlying basement, which includes preexisting ore occurrences.

Keywords Kupferschiefer · Central African Copperbelt · Dolostone Ore Formation · Trace elements · LA-ICP-MS · Sulfides · Random Forest classification · Machine learning

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Introduction

Trace element analysis by laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) has been widely applied to better understand ore-forming processes and to characterize ore deposit types such as Mississippi Valley type (MVT; Hu et al. 2021; Torró et al. 2023), skarn (Chen et al. 2020), porphyry copper, and epithermal (Franchini et al. 2015; Cook et al. 2016; Sykora et al. 2018; Benites et al. 2021; Velićević et al. 2022). The trace element contents of pyrite from sediment-hosted Au deposits (e.g., Large et al. 2007, 2009; Gopon et al. 2019; Xie et al. 2023), volcanic-hosted massive sulfides (VHMS; e.g., González-Jiménez et al. 2022; Torró et al. 2022; Caraballo et al. 2023), and sedimentary pyrite in general (e.g., Large et al. 2014; Gregory et al. 2015) have been widely investigated. Plots of Co versus

Ni in pyrite were proposed by Campbell and Ethier (1984) to discriminate pyrite from sedimentary, volcanic (i.e., hydrothermal), and magmatic affinities. Duran et al. (2015) used Se/As vs Co/Sb to distinguish hydrothermal and magmatic pyrite and Duran et al. (2019) proposed a ternary Se–Ni–Cd discrimination diagram that separates chalcopyrite from hydrothermal and magmatic origins. Caraballo et al. (2023) used Random Forest classification to distinguish chalcopyrite from different VHMS environments. Sphalerite discrimination plots have also been suggested to discriminate different deposit types (Hu et al. 2021; Liu et al. 2022).

Stratiform sediment-hosted Cu and Cu–Co deposits have, until recently, been absent from these studies. This point was recognized by Marsh et al. (2016), who concluded that there are “virtually no data on Ga concentrations” and “very little data” for Ge from stratiform sediment-hosted Cu and Cu–Co deposits. Therefore, this study presents new LA-ICP-MS data for base metal sulfides (chalcopyrite, sphalerite, and pyrite) using LA-ICP-MS from three stratiform sediment-hosted Cu–Co districts: the Polish Kupferschiefer, the Central African Copperbelt (with samples from both the Congolese and Zambian segments), and the Dolostone Ore Formation deposit in Namibia. We use Random Forest (Breiman 2001) to classify the trace element datasets of chalcopyrite, sphalerite, and pyrite from the stratiform sediment-hosted Cu–Co districts to constrain trace element signatures that are distinctive of different paragenetic stages. We compare our results to previous data for stratiform sediment-hosted Cu–Co deposits (Foltyn et al. 2022; Bertrandsson Erlandsson et al. 2022) and discuss the causes for the variation in sulfide chemistry with a focus on the impact of varying metal sources at the district scale.

Geologic setting

The Polish Kupferschiefer

The Lubin–Głogów copper district in southwestern Poland (ESM 1, Figs. S1–S2) is the classic example of the reduced-facies type (i.e., Kupferschiefer type) of the stratiform sediment-hosted Cu deposit-type (e.g., Borg et al. 2012 for general overview). Post-Variscan rift-related tectonics and associated magmatic activity resulted in deposition of the Cisuralian Lower Rotliegend Group rocks composed of red-colored clastic sedimentary rocks and bimodal volcanic rocks. The overlying Upper Rotliegend Group rocks are terrestrial rocks composed of aeolian sandstone interbedded with deposits of alluvial fans, braided rivers, and playas (Karnkowski 1999; Kiersnowski and Buniak 2006; McCann et al. 2006).

In a typical profile, continental red beds end with light grey Weissliegend sandstone, aeolian at the bottom and

transgressive in the upper part, covered by a thin (average 0.3 m thick) layer of black, marine shale overlain by the Zechstein Group limestone (ESM 1 Fig. S2). The Cu–Ag sulfides occur mainly as veinlets, dispersed disseminations, impregnations, coarse-grained aggregate nests and semi-massive ores not only restricted to the Kupferschiefer layer, but are also found in the underlying sandstones and overlying carbonates. Geological resources of the active Cu mines operated by KGHM in the Fore-Sudetic Homocline are 1,487.46 Mt, with average grades of 1.81% Cu and 52 g/t Ag. The amount of metals in the resource estimates is over 56.9 Mt metallic Cu and over 164,000 t of metallic Ag (PGI 2024). New discoveries of Cu deposits in the deeper part of the basin could increase these resources (Speczik et al. 2021; Bieńko and Pietrzela 2022).

Most authors agree that mineralization was protracted and took place during several stages (Wodzicki and Piestrzyński 1994; Michalik and Sawłowicz 2001; Blundell et al. 2003; Alderton et al. 2016). The main mineralizing process was the migration of low-temperature, oxidizing, metalliferous chloride brines likely sourced from paleo-depressions filled with Rotliegend Group red bed sediments and from basement rocks (Blundell et al. 2003), through the anoxic basal sedimentary rocks of the Zechstein Group (Jowett 1986; Oszczepalski 1999). Temperature estimates for the German Kupferschiefer suggest that the shale underwent temperatures up to 130 °C (Bechtel et al. 1995). Speczik and Puttmann (1987) proposed that the mineralizing fluids sporadically exceeded 100 °C and that the highest recorded temperatures are the results of maturation events that post-date the main mineralization stage of the Polish Kupferschiefer. Fluid inclusions from epigenetic veins show homogenization temperatures between 145 and 200 °C (Strengel-Martinez et al. 1993).

The Central African Copperbelt

The world-class Cu–Co deposits of the Central African Copperbelt have a combined total pre-mining endowment of 166 Mt Cu (Taylor et al. 2013) with several deposits having average grades of 5% Cu and 1% Co (ESM1, Figs. S3; Hitzman et al. 2005). They occur in a belt stretching over 400 km with a width of 100 km across the DR Congo and Zambian border. The deposits are hosted in Neoproterozoic sedimentary rocks with an age between 880 and 542 Ma (Armstrong et al. 2005; Master et al. 2005; Cailteux et al. 2018). The host-rocks overly a basement of metasedimentary and magmatic rocks (Van Wilderode et al. 2015). The maximum temperature reached during burial ranges between 220 °C in the Tenke Fungurume mining district in the west (e.g., the Kwabwala deposit, DRC) to 240 °C at the Nkana–Mindola deposit (Zambia) in the east (Liu et al. 2024). The Neoproterozoic sedimentary rocks were deformed during the Pan-African Lufilian orogeny between 560 and 530 Ma

(Hanson et al. 1993; John et al. 2004). Although Cu and Co mineralization occurred along many stratigraphic levels in the Neoproterozoic Roan, Nguba, and Kundelungu groups (ESM 1, Fig. S4; Mambwe et al. 2023), the majority of the economic occurrences are concentrated in the lower part of the Roan Group (Cailteux et al. 2005). The deposits are both stratiform and stratabound (Cailteux et al. 2005; Dewaele et al. 2006). Numerous models have been proposed for the formation of the Central African Copperbelt, ranging from magmatic to syn-sedimentary, diagenetic, and orogenic simple or multiphase origin (Cailteux et al. 2005; Selley et al. 2005). Available data demonstrate that mineralization occurred over a protracted period that spanned diagenesis, basin inversion, and metamorphism (Hitzman et al. 2012).

Luiswishi deposit, DR Congo

The Luiswishi Cu-Co ore deposit is located in the eastern portion of the Congolese part of the Central African Copperbelt (the Katanga Copperbelt) (ESM 1; Fig. S3). Resources at Luiswishi are 8 Mt with 200 kt Cu @ 2.5% and 88 000 t Co @ 1.1% (Taylor et al. 2013). This deposit was selected for our study because it has been thoroughly described and studied (El Desouky et al. 2010 and references therein) and it displays the two main mineralization stages (diagenetic and orogenic) recognized in the Central African Copperbelt. The ore at Luiswishi is concentrated in two formations of the Roan Group. The Lower and Upper orebodies occur in the Kamoto and Kambove formations, respectively (Cailteux et al. 2005). Sulfides from the first mineralization stage are disseminated in the host rock, in nodules and lenses together with gangue minerals such as quartz and dolomite (El Desouky et al. 2010). Analysis of fluid inclusions in these gangue minerals indicates that the mineralizing fluid had intermediate salinities (5–12 eq. wt% NaCl) and temperatures between 115 and 220 °C (El Desouky et al. 2009). Sulfides from the second stage occur in nodules, lenses, veins, and breccia. They precipitated from a high salinity (35–42 eq. wt% NaCl) and high temperature (> 290 °C) fluid (El Desouky et al. 2009).

Nchanga deposit, Zambia

The Nchanga deposit is located in the Zambian part of the Central African Copperbelt (ESM 1, Fig. S3) and has a resource of > 425 Mt @ 3.24% Cu and 0.03% Co, amounting to > 13 Mt of Cu and > 130 000 t of Co (Taylor et al. 2013). Mineralization formed as a result of basin inversion during the Lufilian orogeny (Molak 1995; McGowan et al. 2006) and is therefore attributed to the second main stage of mineralization in the Copperbelt. The Lower Orebody and an Upper Orebody are present in the Lower Roan Subgroup. Sulfide occurrences are mainly disseminated,

breccia-hosted, fracture-related, and vein-hosted and present in coarse-grained quartz-feldspathic rocks (McGowan et al. 2003). The Roan sedimentary rocks are intensely folded and the limbs of the tight folds are in many places overprinted by a thrust (Swardt 1962). Metal-bearing fluids could have derived from the deepest part of the basin and from the basement, and the sulfides formed by thermochemical sulfate reduction at the deposition site (McGowan et al. 2003). The mineralizing fluid had a high salinity (31–38 eq. wt% NaCl) and a minimum temperature of 140 – 180 °C (McGowan et al. 2006).

The Namibian Dolostone Ore Formation deposit

The “Dolostone Ore Formation” is the given name of the Cu-Co-Zn deposit, and does not represent a “formation” in a stratigraphic sense. The deposit currently has indicated plus inferred mineral resources of 225.5 Mt @ 0.12%, Co 0.43% Cu, and 0.54% Zn (Celsius Resources Limited 2021). Fine-grained siliciclastic and carbonate rocks of the Ombombo Subgroup in the Neoproterozoic lower Damara Supergroup are the main host rocks (ESM 1, Figs. S5–S6; Allen 2016). The Damara Supergroup is a rift to carbonate platform succession with basal fluvial clastic red bed sedimentary rocks (Nosib Group) that are predominately composed of arkosic to feldspathic sandstone deposited in half grabens (Guj 1970; Porada 1989; Miller 2008; Allen 2016). The Otavi Group overlies the Nosib Group and comprises cyclical marine carbonate and siliciclastic rocks, with an upward increasing carbonate component. In the area of the Dolostone Ore Formation deposit, the Otavi group is composed of carbonate, calcareous shale, and siltstone (Hoffman and Halverson 2008; Miller 2008; Allen 2016). The deposit is situated within the Eastern Foreland Zone of the NE-SW trending Kaoko Belt (Goscombe et al. 2005). The eastern foreland experienced greenschist-facies regional metamorphism during the Damaran Orogeny that resulted in folding and local shearing (Goscombe 1999; Goscombe et al. 2003, 2005). Temperature estimates of the regional metamorphism in the Dolostone Ore Formation area range between 300 and 400 °C (Guj 1970; Foster et al. 2009).

Sulfides occur as disseminations, polysulfide nodules, and aggregates within the host rock along with vein-hosted and pressure-shadow mineralization. Two main ore-forming stages have been proposed: (1) syn-metamorphic Cu-Co-Zn and (2) late Cu–Zn(-Pb) vein and pressure shadow mineralization. The sulfide assemblage is dominantly composed of chalcopyrite, sphalerite, and pyrite with subordinate linnaeite and pyrrhotite and minor to accessory galena and cobaltite (Bertrandsson Erlandsson et al. 2022). Bertrandsson Erlandsson et al. (2022) attribute the first mineralization stage to the Damaran Orogeny and the second mineralization stage to post orogenesis.

Samples and analytical methods

Sampled deposits

Data from thirty-five samples from four stratiform sediment-hosted Cu(-Co) districts (Table 1) are included in this study: eleven samples from the Polish Kupferschiefer; five from the Congolese Luiswishi deposit (Katanga Copperbelt) and two from the Nchanga deposit (Zambian Copperbelt) of the Central African Copperbelt; and fifteen samples from the Namibian Dolostone Ore Formation deposit. Data are limited for Nchanga due to access to samples, and therefore although the data are included for comparison, they were omitted from Random Forest analyses. Our results were combined with some previously published data for the Polish Kupferschiefer and Nchanga deposits (Foltyn et al. 2022) and the Namibian Dolostone Ore Formation deposit (Bertrandsson Erlandsson et al. 2022). All data are listed in ESM 2 (Tables S1–S3).

Samples were subdivided into two main ore stages based on previously published petrographic evidence (Polish Kupferschiefer from Oszczepalski (1999) and Foltyn et al. (2022); Katanga Copperbelt from El Desouky et al. (2010); and Dolostone Ore Formation deposit from Bertrandsson Erlandsson et al. (2022). Petrographic and paragenetic summaries of the investigated samples and deposits are listed in ESM 3, Figs. S7–S13. Sulfides deposited during stage 1 occur primarily as dissemination and mineralized nodules (e.g., stratiform mineralization-style). Sulfides deposited during stage 2 are predominantly in veins that cross-cut sedimentary bedding.

Stage 1 mineralization typically formed during diagenesis, whereas stage 2 is related to later hydrothermal events. In the Dolostone Ore Formation deposit, certain gangue-dominated veins (referred to as “DOF events”) are related to the first main Cu-Co stage (Bertrandsson Erlandsson et al. 2022). The simplification of two main mineralization stages is predominantly made based on the mode of occurrence and petrographic evidence presented in the aforementioned publications. Precise age determination of individual mineralization stages of stratiform sediment-hosted Cu(-Co) deposits is challenging, due to the prolonged ore-forming process involved in ore formation (e.g., Bartholomé et al. 1972; Jowett et al. 1987; Bechtel et al. 1999; Cailteux et al. 2005; Brems et al. 2009; El Desouky et al. 2010; Symons et al. 2011; Hitzman et al. 2012; Alderton et al. 2016; Cailteux et al. 2018).

Laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS)

LA-ICP-MS trace element analyses were carried out at the Montanuniversität Leoben, Austria, using an ESI NWR213 Nd:YAG laser ablation system coupled to an Agilent 8800 QQQ ICP-MS. This makes the novel and previously published data directly comparable as the same instrument set up and analytical conditions were used. A 50 µm spot size was used with an on-mineral fluency of 2–3 J/cm² for the sulfides with a repetition rate of 10 Hz. Helium carrier gas was used at a 0.75 L/min flow rate in the laser ablation cell. Background values are collected during a 30-s pre-ablation laser warm-up, which is followed by 60 s dwell time when

Table 1 Genetic parameters of stratiform sediment-hosted copper deposits investigated in this study

District		Age	Ore stages	Formation temperature
	Deposit			
Polish Kupferschiefer		Guadalupian—Early Triassic ^a	1	80 – 140 °C ^{b,c}
		Triassic – Jurassic (?) ^d	2	100 – 200 °C ^{b,e}
Central African Copperbelt				
DRC	Luiswishi	816±62 Ma ^f	1	115 – 220 °C ^g
		592–512 Ma ^h	2	270 – 385 °C ^g
Zambia	Nchanga	~ 583±24 Ma ⁱ	2	105 – 300 °C ^j
Kunene Namibia	DOF	~ 520—480 Ma	1	> 310 – 400 °C ^{k,l,m}
		(uncertain) ^k	2	~ 310 °C ^k

Ore Stage 1 refers to sediment-hosted mineralization, related to diagenesis. Ore stage 2 refers to mineralization within epigenetic hydrothermal veins, typically orogenic (with the exception of the Polish Kupferschiefer), and accompanied by abundant gangue minerals such as quartz and carbonates. All formation temperatures are based on fluid inclusion data, except for the Dolostone Ore Formation deposit in which case are derived from sphalerite trace element content and regional metamorphic temperatures, and Kupferschiefer stage 1 that is estimated based on stable isotopes and organic matter maturation. References may not always strictly relate to the investigated deposit, but rather the metallogenic district. Abbreviations: DOF=Dolostone Ore Formation, DRC=Democratic Republic of Congo. References: a=Alderton et al. (2016); b=Borg et al. (2012); c=Vaughan et al. (1989); d=Jowett (1987); e=Strengel-Martinez et al. (1993); f=Selley et al. (2005); g=El Desouky et al. (2009); h=Rinaud et al. (2005); i=Barra et al. (2004); j=McGowan et al. (2006); k=Bertrandsson Erlandsson et al. (2022); l=Foster et al. (2009); m=Guj (1970)

the laser is active. Each laser ablation cycle is separated by a 30-s washout period.

Analyzed isotopes were: ^{34}S , ^{51}V , ^{52}Cr , ^{55}Mn , ^{57}Fe , ^{59}Co , ^{60}Ni , ^{63}Cu , ^{67}Zn , ^{71}Ga , ^{72}Ge , ^{73}Ge , ^{74}Ge , ^{75}As , ^{82}Se , ^{95}Mo , ^{107}Ag , ^{111}Cd , ^{115}In , ^{118}Sn , ^{121}Sb , ^{125}Te , ^{197}Au , ^{201}Hg , ^{205}Tl , ^{208}Pb , and ^{209}Bi . Belissont et al. (2014) demonstrated that ^{74}Ge is the most suitable germanium mass for sulfide (sphalerite) analyses, due to the potential polyatomic interference of As-S species on all germanium isotopes. Nevertheless, ^{72}Ge and ^{73}Ge were added in the element suite of this study to monitor Ar-S interferences. The software Iolite 4 (Paton et al. 2011) was utilized for data reduction. Detection limits were calculated using the “Howell method” in the Iolite 4 software. Additionally, ^{24}Mg , ^{27}Al , ^{28}Si , and ^{43}Ca were monitored during LA-ICP-MS analyses, but not quantified to determine the presence of gangue mineral inclusions within the measured sulfides and potential element associations to the gangue inclusions. Spot measurements with obvious inclusions were removed from the dataset, as this study focuses on the trace element content of the targeted sulfides.

The USGS pressed powder polysulfide reference material MASS-1 (Wilson et al. 2002) was used as external standard for data quantification of chalcopyrite, with stoichiometric Cu as the internal standard. For sphalerite and pyrite, the sphalerite matrix-matched sinter pressed powder pellet reference material MUL-ZnS1 (Onuk et al. 2017) was used as external standard, with stoichiometric S as internal standard. With the exception of the quantification of Te, Au, Tl, and Hg in sphalerite and pyrite, the MASS-1 was used as the external standard for these elements as the MUL-ZnS1 is not suitable for quantification of these elements. To correct for instrumental drift and quality control during the analytical runs, both reference materials were regularly analyzed after every 20 spots of unknowns. Sulfide minerals may exhibit heterogeneous trace element distributions (Cook et al. 2016), and thus 20 spot analyses per sulfide phase per sample was sought after to get a representative dataset. Naturally, this is not always possible due to limiting factors such as amount and size of sulfide phases. All trace element contents herein are reported as $\mu\text{g/g}$, if not noted otherwise.

Random forest classification

Binary Random Forest analyses (Breiman 2001) were applied to the trace element datasets to investigate element signatures of the different stratiform sediment-hosted Cu(-Co) districts. The Random Forest analysis ranks each element on how well it distinguishes the two datasets, indicated by the Gini value. The Gini value works as an index to define the best split threshold in the Random Forest decision trees at each node generated (Gregory et al. 2019). The higher the Gini value of an element is relative to another, the more suitable the element is at discriminating the two datasets.

Random Forest statistical evaluations were carried out in R (Development Core Team 2013) using the Random Forest package.

Fifty spot measurements were used for comparison to the Kupferschiefer dataset and 70 for the Central African Copperbelt-Dolostone Ore Formation. A minimum of 29 data points had to be used when carrying out the Random Forest analyses due to varying population size of the datasets. Median values of each element of the separate datasets were inserted into missing cells. Samples are picked at random to avoid overfitting. Each Random Forest dataset comparison was carried out five times to evaluate the impact of the random sample selection; from this, a mean Gini value was produced. Five hundred trees were generated each run with four variables tied at each of the splits. Random Forest analyses were carried out on sulfides from the Polish Kupferschiefer, Katanga Copperbelt, and Namibian Dolostone Ore Formation deposit. Additionally, the trace element datasets for the Katanga Copperbelt and Dolostone Ore Formation sulfides were separated into the two main ore forming stages. This was not done for the Polish Kupferschiefer, as the dataset is not large enough to make such a division without overfitting. The dataset for the Zambian Copperbelt deposit was too small to be included in the Random Forest analyses.

Results

Results of sulfide mineral chemistry must account for the presence of gangue minerals or other sulfide minerals that are intergrown with or occur as inclusions within the sulfide minerals. We removed data points that represent obvious inclusions, resulting in a high variability of number of data points per element in a sample set (ESM 2, Tables S1-S3). Vanadium and Cr, for example, are commonly correlated with Si and Al count signals, attributed to Al-bearing silicates from the host rock that occur as inclusions within the sulfides (ESM 4, Fig. S14). Correlation between Mn and Ca is also common. The Ca is attributed to carbonate inclusions, as carbonates are common gangue minerals in these deposits, explaining higher Mn values in the dataset.

Sulfide trace element contents of stratiform sediment-hosted Cu(-Co) deposits

Chalcopyrite

A summary of chalcopyrite compositions is presented in Table 2 and Fig. 1. The complete chalcopyrite trace element dataset can be found in ESM 2, Table S1. Median contents of Co are mostly at the single-digit $\mu\text{g/g}$ level, except for those for stage 2 from the Namibian Dolostone Ore Formation deposit at 0.17 $\mu\text{g/g}$. Stage 1 from

Table 2 Statistical summary of trace element contents in chalcopyrite from the Polish Kupferschiefer, Katanga Copperbelt, and Dolostone Ore Formation. All values are presented in $\mu\text{g/g}$. Abbreviations: BDL=below detection limit; DL=detection limit; N=number of data points; n=number of values below the detection limit

	Polish Kupferschiefer													
	Stage 1							Stage 2						
	Median	Min	Max	N	n	DL		Median	Min	Max	N	n	DL	
V	14	0.71	1394	29	0	0.079	BDL	BDL	BDL	BDL	46	39	0.071	
Cr	-	-	-	na	na	-	-	-	-	-	na	na	-	
Mn	24	14	640	29	0	0.43	13	3.9	51	48	0	0.42		
Co	0.57	0.15	49	29	0	0.055	3.7	BDL	33	46	5	0.054		
Ni	9.7	1.5	210	29	0	0.27	BDL	BDL	3.7	42	35	0.24		
Zn	BDL	BDL	50	26	22	13	50.1	BDL	4614	47	10	15		
Ga	0.32	BDL	14	28	5	0.063	0.64	BDL	4.5	48	6	0.051		
Ge	0.42	BDL	7.8	29	4	0.17	1240	0.19	4806	48	0	0.18		
As	BDL	BDL	36	27	15	1.9	307	BDL	1045	45	5	1.6		
Se	36	15	69	29	0	9.3	40	BDL	192	46	11	9.4		
Mo	5.7	BDL	828	29	6	0.082	0.077	BDL	3.0	47	26	0.064		
Ag	57	8.8	773	29	0	0.017	14	2.3	500	48	0	0.014		
Cd	BDL	BDL	BDL	25	25	1.8	BDL	BDL	99	44	36	1.7		
In	1.5	0.019	2.8	27	0	0.020	0.37	0.021	222	48	0	0.016		
Sn	1.1	0.30	9.4	29	0	0.23	1.6	BDL	544	47	1	0.18		
Sb	1.8	0.75	18	29	0	0.082	30	BDL	395	47	5	0.087		
Te	BDL	BDL	1.1	29	21	0.18	BDL	BDL	1.5	47	43	0.21		
Au	BDL	BDL	0.029	29	27	0.028	BDL	BDL	0.31	47	39	0.028		
Hg	7.9	BDL	59	29	13	4.9	4.2	BDL	121	48	13	1.5		
Tl	1.5	0.40	13	29	0	0.017	0.60	BDL	1.9	48	1	0.018		
Pb	137	46	1614	29	0	0.068	152	18	4441	46	0	0.079		
Bi	0.054	BDL	2.8	26	3	0.018	0.10	BDL	4784	46	4	0.014		
Katanga Copperbelt														
	Stage 1							Stage 2						
	Median	Min	Max	N	n	DL		Median	Min	Max	N	n	DL	
V	BDL	BDL	20	42	24	0.058	BDL	BDL	0.49	38	36	0.044		
Cr	BDL	BDL	26	48	37	2.7	BDL	BDL	BDL	42	42	1.3		
Mn	BDL	BDL	33	40	21	0.48	BDL	BDL	22	33	32	0.34		
Co	2.9	1.5	10	44	0	0.13	2.7	1.9	11	43	0	0.093		
Ni	BDL	BDL	1.8	43	33	0.64	BDL	BDL	BDL	41	41	0.49		
Zn	BDL	BDL	95	46	30	16	BDL	BDL	BDL	37	37	11		
Ga	19	7.4	38	53	0	0.15	70	36	86	43	0	0.10		
Ge	6.4	4.5	12	53	0	0.32	3.8	3.0	9.6	43	0	0.25		
As	BDL	BDL	5.9	51	30	1.0	BDL	BDL	BDL	43	43	0.86		
Se	31	BDL	45	53	1	13	13	BDL	23	43	14	8.9		
Mo	0.66	BDL	8.3	44	31	0.17	BDL	BDL	BDL	43	43	0.13		
Ag	BDL	BDL	0.47	45	26	0.057	0.37	BDL	3.1	37	18	0.044		
Cd	BDL	BDL	BDL	52	52	7.2	BDL	BDL	BDL	43	43	9.6		
In	19	2.5	43	53	0	0.040	31	24	39	43	0	0.037		
Sn	50	35	157	53	0	1.6	49	43	64	43	0	1.3		
Sb	0.44	BDL	126	52	16	0.17	BDL	BDL	1.5	41	40	0.14		
Te	BDL	BDL	BDL	50	50	4.8	BDL	BDL	BDL	43	43	4.4		
Au	BDL	BDL	1.9	53	29	0.23	BDL	BDL	BDL	40	40	0.023		
Hg	2.4	BDL	6.2	53	6	1.2	1.6	BDL	5.5	42	7	1.0		
Tl	0.055	BDL	1.2	52	24	0.030	BDL	BDL	BDL	41	41	0.030		
Pb	1.8	BDL	67	48	19	0.10	BDL	BDL	0.48	30	23	0.067		
Bi	25	BDL	222	44	8	0.031	0.088	0.0073	0.86	32	0	0.018		

Table 2 (continued)

Zambian Copperbelt												
Stage 2												
	Median	Min	Max	N	n	DL						
V	BDL	BDL	1.2	18	15	0.058						
Cr	-	-	-	na	na	-						
Mn	14	14	20	18	0	0.35						
Co	3.7	2.0	71	18	0	0.035						
Ni	BDL	BDL	10	18	16	0.21						
Zn	222	BDL	2273	17	5	12						
Ga	3.5	0.26	11	18	0	0.035						
Ge	0.80	BDL	2.0	18	3	0.15						
As	BDL	BDL	BDL	18	18	1.8						
Se	BDL	BDL	19	18	9	8.1						
Mo	BDL	BDL	0.24	18	16	0.060						
Ag	0.62	0.26	2.6	17	0	0.012						
Cd	BDL	BDL	3.3	17	10	1.4						
In	5.2	3.7	5.7	18	0	0.019						
Sn	32	14	70	18	0	0.22						
Sb	BDL	BDL	0.32	18	10	0.077						
Te	0.20	BDL	0.41	18	9	0.17						
Au	BDL	BDL	0.057	18	12	0.014						
Hg	5.2	BDL	26	18	3	1.6						
Tl	BDL	BDL	0.021	18	12	0.010						
Pb	0.76	0.13	9.7	17	0	0.044						
Bi	0.70	0.25	2.4	15	0	0.017						
Dolostone Ore Formation												
Stage 1							Stage 2					
	Median	Min	Max	N	n	DL	Median	Min	Max	N	n	DL
V	BDL	BDL	2.7	132	109	0.074	BDL	BDL	0.15	47	45	0.098
Cr	BDL	BDL	10	168	157	0.95	BDL	BDL	BDL	47	47	0.74
Mn	4.2	BDL	22	126	43	0.50	4.4	3.8	5.2	47	0	0.60
Co	8.6	BDL	164	138	1	0.091	0.17	BDL	0.74	47	11	0.080
Ni	0.66	BDL	62	156	57	0.47	54	BDL	62	47	1	0.40
Zn	339	112	821	149	0	25	137	45	514	36	0	31
Ga	0.22	BDL	3.8	172	30	0.080	0.90	0.37	1.6	46	0	0.061
Ge	0.41	BDL	7.0	178	44	0.24	0.80	0.44	1.6	47	0	0.23
As	BDL	BDL	BDL	188	188	1.3	BDL	BDL	BDL	46	46	0.90
Se	23	BDL	135	177	20	13	104	73	137	47	0	14
Mo	BDL	BDL	2.2	185	184	0.12	BDL	BDL	1.1	47	40	0.096
Ag	8.3	BDL	35	195	7	0.039	3.8	0.13	6.4	47	0	0.032
Cd	BDL	BDL	5.1	192	190	4.1	BDL	BDL	11	46	33	5.5
In	6.5	0.097	70	195	0	0.030	4.3	1.6	31	47	0	0.016
Sn	18	0.44	131	194	0	1.1	59	1.3	95	47	0	0.50
Sb	BDL	BDL	1.0	187	163	0.16	BDL	BDL	1.5	47	30	0.12
Te	BDL	BDL	7.1	193	127	0.40	BDL	BDL	1.0	47	44	0.36
Au	BDL	BDL	0.31	195	157	0.040	BDL	BDL	BDL	47	47	0.032
Hg	5.2	BDL	42	189	17	1.3	3.2	BDL	12	47	1	1.2
Tl	BDL	BDL	2.5	161	122	0.030	BDL	BDL	0.053	47	47	0.028
Pb	5.7	BDL	138	164	1	0.077	0.89	0.11	31	46	0	0.067
Bi	0.65	BDL	24	174	3	0.021	0.028	BDL	0.27	47	15	0.017

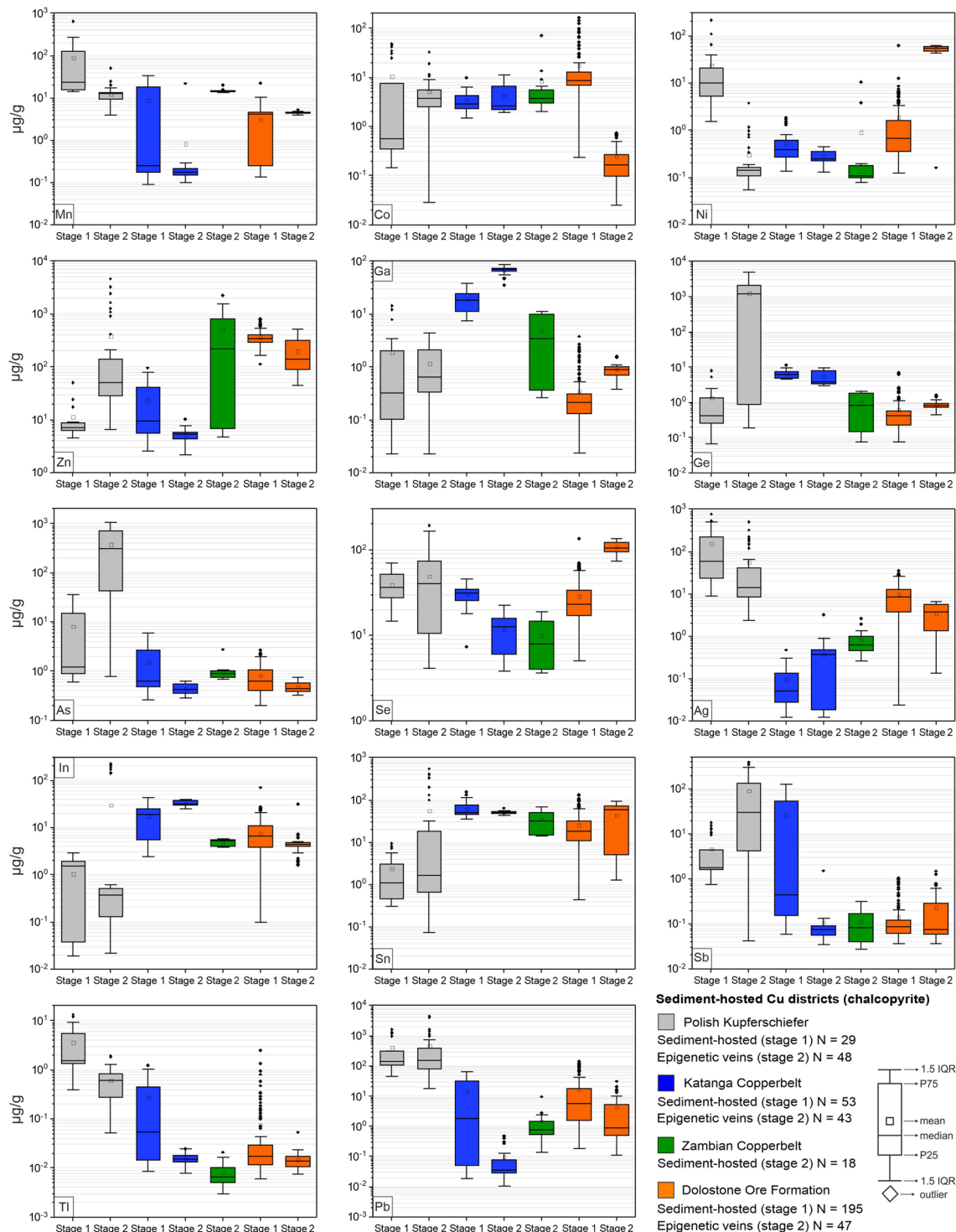


Fig. 1 Chalcopyrite trace element geochemistry of selected elements between the four investigated chalcopyrite-bearing deposits. Data for the Polish Kupferschiefer and Zambian Copperbelt is from Foltyn

et al. (2022); the Dolostone Ore Formation is from this study and Bertrandsson Erlandsson et al. (2022). N = number of LA-ICP-MS analyses for each group

the Polish Kupferschiefer and stage 2 from the Namibian Dolostone Ore Formation deposit have median Ni values in the tens of $\mu\text{g/g}$ compared to Ni in the other

chalcopyrite groups (median values $< 1 \mu\text{g/g}$). Median Zn contents (hundreds of $\mu\text{g/g}$) in the Dolostone Ore Formation and Zambian Copperbelt deposits are higher than the

other chalcopyrite groups. Both Ga and Ge are elevated in Katanga Copperbelt chalcopyrite relative to others, with Ga contents higher than Ge. Stage 2 from the Polish Kupferschiefer has median Ge contents (1,240 µg/g) that are significantly higher than all other groups with median Ge contents of around 1 µg/g. Selenium contents are relatively homogeneously distributed, with only stage 2 from the Dolostone Ore Formation deposit exhibiting slightly higher median Se contents (104 µg/g). Silver is highest in chalcopyrite from the Polish Kupferschiefer (median tens of µg/g), whereas the other groups have Ag contents just above or below 1 µg/g. Chalcopyrite from the African stratiform sediment-hosted districts displays higher median Sn contents (tens of µg/g) compared to the Polish Kupferschiefer (just above 1 µg/g). Median contents of Sb, Tl, and Pb show similar trends of being elevated in the Polish Kupferschiefer compared to the other chalcopyrite groups.

Sphalerite

Sphalerite was only identified in samples from the Polish Kupferschiefer and the Dolostone Ore Formation deposit. A summary of sphalerite compositions is presented in Table 3 and Fig. 2. The complete sphalerite trace element dataset can be found in ESM 2, Table S2. Trace element contents vary significantly between them and between different mineralization stages. Both Mn and Fe contents are higher in Dolostone Ore Formation than the Kupferschiefer and within each deposit, stage 1 sphalerite is higher than stage 2 (Fig. 2). Stage 1 in the Dolostone Ore Formation has the highest median Mn content (201 µg/g), whereas stage 2 sphalerite of the Kupferschiefer contains the lowest (median < 0.34 µg/g). Iron contents (median of 10,000 µg/g) in Dolostone Ore Formation are significantly higher than both stage 1 (1502 µg/g) and stage 2 (125 µg/g) of the Kupferschiefer sphalerite. Cobalt contents are also higher in the Dolostone Ore Formation (particularly stage 1) compared to Kupferschiefer samples (median in single digit µg/g range).

Gallium is only elevated in stage 1 sphalerite of the Polish Kupferschiefer (median 18 µg/g), whereas the other groups have median Ga contents of 1–2 µg/g or below. Germanium is higher in the Kupferschiefer sphalerite (single-digit µg/g range), compared to the Dolostone Ore Formation with Ge contents < 1 µg/g. Median silver contents are the highest in stage 1 of the Kupferschiefer (33 µg/g). In contrast, stage 2 sphalerite of the Kupferschiefer and the Dolostone Ore Formation has a median Ag content in the single-digit µg/g range. The median content of Cd is the highest in the Kupferschiefer stage 2 sphalerite (9175 µg/g). Stage 1 Dolostone Ore Formation sphalerite has median Cd contents below the thousands of µg/g range (222 µg/g). The median content of

In is much higher in the Dolostone Ore Formation sphalerite (tens of µg/g range), compared to Kupferschiefer which has median In contents < 1 µg/g. Median Tl contents are elevated in sphalerite from the Kupferschiefer with contents in the single-digit µg/g range, whereas the Dolostone Ore Formation sphalerite has median Tl contents < 1 µg/g.

Pyrite

A summary of pyrite compositions is presented in Table 4 and Fig. 3. The complete pyrite trace element dataset can be found in ESM 2, Table S3. Stage 1 Polish Kupferschiefer pyrite has the highest median Mn content (240 µg/g) compared to the other pyrite groups. Median Mn contents are elevated in stage 1 compared to stage 2 in both the Katanga Copperbelt and the Dolostone Ore Formation. Median Co contents are significantly higher in the Katanga Copperbelt and Dolostone Ore Formation (thousands to ten-thousand µg/g range), compared to the Kupferschiefer (median 91 µg/g). Katanga Copperbelt stage 2 has the highest median Co content (77,058 µg/g). Pyrite from the Katanga Copperbelt and Dolostone Ore Formation have higher median Ni contents (hundreds µg/g range), compared to the Polish Kupferschiefer (median Ni 25 µg/g). Copper is the highest in the Polish Kupferschiefer stage 1 pyrite with a median Cu of 4,593 µg/g, followed by Katanga Copperbelt stage 1 (median 564 µg/g). The other groups have median Cu contents in the tens of µg/g range or less. Median As contents are highest in Kupferschiefer and the Katanga Copperbelt pyrite (hundreds of µg/g range). Thallium is significantly higher in the Kupferschiefer pyrite (median 846 µg/g), as the other pyrite groups have median Tl contents < 1 µg/g – only stage 1 Katanga Copperbelt has a median Tl content in the single-digit µg/g range. Polish Kupferschiefer stage 1 pyrite has the highest median Pb content (1,706 µg/g), compared to the other groups with median Pb in the single-digit µg/g range or below.

Random forest analysis

Chalcopyrite is the only ore mineral that occurs in all of the investigated metallogenic districts. However, chalcopyrite from the Nchanga deposit (Zambian Copperbelt) had to be excluded from the Random Forest statistics because the sample population is too low. Random Forest analyses were done in pairs between the three remaining deposits (ESM 5, Fig. S16). Thallium, Sb, Pb, Mn, Zn, As, and In (listed from suitable to less suitable) are the best discriminating elements for chalcopyrite between the Polish Kupferschiefer and the Dolostone Ore Formation. Gallium, Ge, Zn, Ag, S, In, Ni, and Pb (listed from more to less suitable) are the best discriminating elements for chalcopyrite between the Katanga Copperbelt and the Dolostone Ore Formation. Sphalerite

Table 3 Statistical summary of trace element contents in sphalerite from the Polish Kupferschiefer and Dolostone Ore Formation. All values are presented in $\mu\text{g/g}$. Abbreviations: BDL=below detection limit; DL=detection limit; N=number of data points; n=number of values below the detection limit

	Polish Kupferschiefer						Stage 2					
	Median	Min	Max	N	n	DL	Median	Min	Max	N	n	DL
V	0.60	0.22	47	11	0	0.046	BDL	BDL	0.13	32	23	0.059
Cr	BDL	BDL	12	15	9	0.93	BDL	BDL	1.7	32	10	0.58
Mn	11	6.0	154	22	0	0.51	BDL	BDL	25	32	22	0.34
Fe	1502	432	7688	32	0	20	125	58	2069	32	0	9.5
Co	BDL	BDL	7.5	28	13	0.13	4.2	0.58	49	32	0	0.051
Ni	3.0	BDL	40	30	11	0.69	BDL	BDL	0.35	32	20	0.22
Cu	2317	BDL	4810	26	1	5.2	266	22	11,095	32	0	1.8
Ga	18	1.5	108	33	0	0.16	0.58	0.053	7.1	32	0	0.032
Ge	3.5	BDL	60	33	1	0.38	1.2	BDL	7.9	32	6	0.17
As	BDL	BDL	19	33	23	1.2	0.69	BDL	16	32	4	0.69
Se	15	8.7	398	29	0	8.6	15	2.3	23	32	0	6.0
Mo	10	BDL	99	33	8	0.16	BDL	BDL	14	32	17	0.071
Ag	33	21	201	33	0	0.082	2.7	0.28	29	32	1	0.023
Cd	4673	1342	8974	33	0	1.9	9175	4540	17,835	32	0	2.1
In	BDL	BDL	0.39	33	20	0.038	0.039	BDL	22	32	3	0.017
Sn	3.6	BDL	11	32	1	1.2	1.2	BDL	208	32	2	0.27
Sb	1.1	BDL	8.9	33	2	0.10	0.20	BDL	2.6	32	14	0.085
Te	BDL	BDL	0.82	33	32	0.54	BDL	BDL	BDL	32	32	0.13
Au	BDL	BDL	BDL	28	28	0.59	BDL	BDL	BDL	18	18	0.35
Hg	130	BDL	225	33	1	2.9	54	9.2	175	32	0	2.7
Tl	2.4	0.73	4.4	33	0	0.067	1.1	BDL	15	32	1	0.019
Pb	102	24	42,681	26	0	0.13	25	BDL	325	32	2	0.24
Bi	BDL	BDL	0.37	31	26	0.022	BDL	BDL	0.075	32	17	0.012

	Dolostone Ore Formation						Stage 2					
	Median	Min	Max	N	n	DL	Median	Min	Max	N	n	DL
V	BDL	BDL	0.18	118	108	0.053	BDL	BDL	BDL	75	75	0.064
Cr	BDL	BDL	0.99	138	135	0.70	BDL	BDL	BDL	75	75	0.63
Mn	201	80	339	142	0	0.39	44	13	230	75	0	0.42
Fe	65,268	38,664	73,529	147	0	17	38,599	14,043	58,622	74	0	21
Co	8672	4311	9496	151	0	0.091	799	32	1637	75	0	0.072
Ni	0.86	BDL	8.5	146	22	0.38	19	1.1	37	74	0	0.40
Cu	104	12	88	85	0	3.0	97	26	974	69	0	2.7
Ga	0.36	BDL	1.3	148	2	0.067	1.5	0.56	7.7	75	0	0.055
Ge	BDL	BDL	0.65	149	118	0.22	0.48	BDL	1.4	75	3	0.21
As	BDL	BDL	8.8	148	83	0.94	BDL	BDL	BDL	74	74	0.74
Se	28	15	32	151	0	8.2	99	53	241	75	0	10
Mo	BDL	BDL	0.65	149	148	0.067	BDL	BDL	BDL	75	75	0.067
Ag	0.60	BDL	1.6	138	5	0.028	0.27	0.049	7.2	75	0	0.026
Cd	222	107	349	151	0	3.5	1865	1285	2550	75	0	3.9
In	34	5.3	11	151	0	0.021	21	17	152	75	0	0.016
Sn	1.4	BDL	2.8	149	42	0.54	BDL	BDL	7.0	75	46	0.45
Sb	BDL	BDL	0.070	150	149	0.11	BDL	BDL	BDL	74	74	0.11
Te	BDL	BDL	2.9	151	146	0.35	BDL	BDL	BDL	75	75	0.31
Au	BDL	BDL	0.38	127	124	0.023	BDL	BDL	BDL	75	75	0.020
Hg	68	29	159	151	0	1.0	100	59	153	75	0	1.0
Tl	BDL	BDL	0.016	150	136	0.021	BDL	BDL	BDL	74	74	0.022
Pb	1.2	BDL	0.82	128	17	0.052	0.11	BDL	5.8	73	19	0.063
Bi	0.13	BDL	0.043	137	26	0.011	BDL	BDL	0.040	73	62	0.013

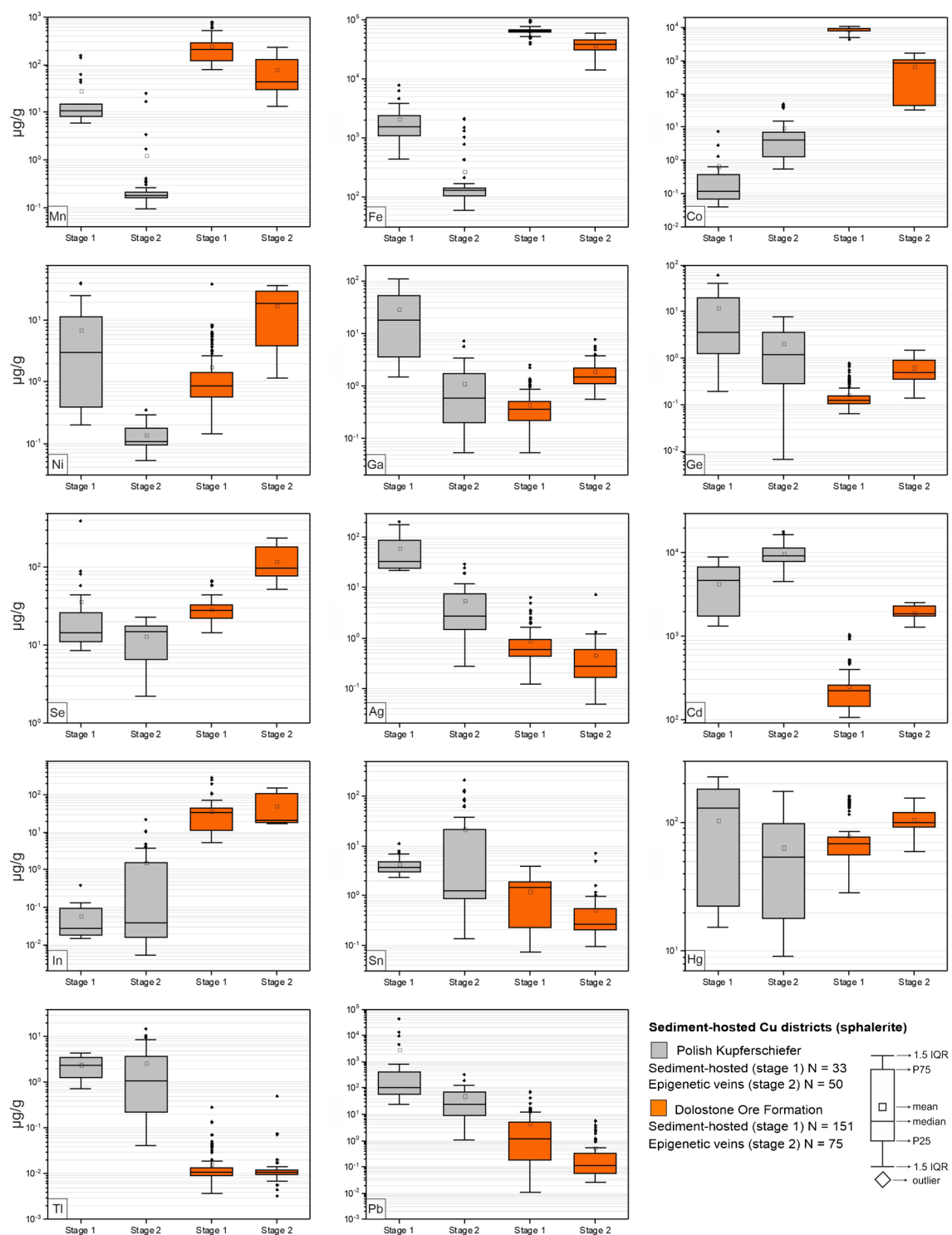


Fig. 2 Sphalerite trace element geochemistry of selected elements between the Polish Kupferschiefer and the Dolostone Ore Formation. Data for the Polish Kupferschiefer is from Foltyn et al. (2022) and

this study; the Dolostone Ore Formation is from this study and Bertrandsson Erlandsson et al. (2022). N=number of LA-ICP-MS analyses for each group

Table 4 Statistical summary of trace element contents in pyrite from the Polish Kupferschiefer, Katanga Copperbelt, and Dolostone Ore Formation. All values are presented in µg/g. Abbreviations:

BDL=below detection limit; DL=detection limit; N=number of data points; n=number of values below the detection limit

Polish Kupferschiefer

Stage 1						
	Median	Min	Max	N	n	DL
V	4.4	1.6	6.5	11	0	0.017
Cr	1.0	0.57	2.0	14	0	0.33
Mn	260	164	787	17	0	0.17
Co	91	14	154	14	0	0.041
Ni	25	12	67	13	0	0.16
Cu	4593	786	7047	8	0	1.1
Zn	33	23	73	18	0	4.0
Ga	BDL	BDL	0.059	8	6	0.051
Ge	0.14	BDL	0.24	14	6	0.12
As	137	64	235	14	0	0.41
Se	10	3.1	17	18	0	2.9
Mo	25	8.7	116	18	0	0.61
Ag	13	2.7	29	7	0	0.016
Cd	19	13	27	18	0	0.67
In	0.095	0.028	0.15	18	0	0.015
Sn	1.5	0.84	1.9	18	0	0.30
Sb	2.4	1.4	4.2	18	0	0.050
Te	bdl	bdl	bdl	18	18	na
Au	BDL	BDL	BDL	18	18	0.17
Hg	2.7	1.7	6.9	18	0	0.85
Tl	846	612	1254	17	0	0.019
Pb	1706	249	2861	18	0	0.063
Bi	0.036	BDL	0.12	18	3	0.0048

Katanga Copperbelt

Stage 1							Stage 2					
	Median	Min	Max	N	n	DL	Median	Min	Max	N	n	DL
V	1.4	0.94	1.8	2	0	0.022	BDL	BDL	BDL	16	14	0.023
Cr	3.2	1.1	4.7	16	0	0.52	BDL	BDL	BDL	29	29	0.38
Mn	7.9	2.7	17	10	0	0.23	0.64	0.30	3.7	11	0	0.21
Co	8480	206	23,153	21	0	0.16	77,058	48,968	108,174	31	0	0.19
Ni	154	37	384	22	0	0.31	277	155	434	30	0	0.31
Cu	564	101	855	8	0	2.8	9.2	2.8	26	16	0	1.1
Zn	11	BDL	106	11	5	8.5	BDL	BDL	BDL	29	29	6.4
Ga	0.61	0.32	1.5	6	0	0.088	BDL	BDL	BDL	18	18	0.070
Ge	1.2	0.61	3.2	7	0	0.21	0.33	BDL	0.79	29	2	0.14
As	367	20	643	10	0	0.89	90	25	198	22	0	0.67
Se	15	8.8	25	18	0	5.8	55	25	140	31	0	4.8
Mo	BDL	BDL	0.67	12	11	0.049	BDL	BDL	BDL	31	31	0.076
Ag	0.16	BDL	0.22	15	6	0.15	BDL	BDL	BDL	31	31	0.034
Cd	BDL	BDL	BDL	18	18	2.2	BDL	BDL	BDL	31	31	1.4
In	0.73	BDL	1.8	15	1	0.024	BDL	BDL	0.034	26	22	0.016
Sn	3.2	BDL	8.7	16	1	0.55	BDL	BDL	0.52	27	27	0.50
Sb	5.3	0.73	13	16	0	0.091	0.065	BDL	0.39	28	9	0.044
Te	BDL	BDL	BDL	18	18	na	bdl	bdl	2.6	31	30	na
Au	BDL	BDL	0.042	18	14	0.015	BDL	BDL	0.14	29	17	0.014
Hg	1.3	BDL	3.7	18	6	0.89	BDL	BDL	BDL	30	30	1.0
Tl	0.061	0.019	0.17	16	0	0.016	BDL	BDL	BDL	31	31	0.010
Pb	11	3.9	46	15	0	0.020	0.034	BDL	0.14	17	4	0.016
Bi	8.6	1.5	16	14	0	0.0066	0.64	0.15	3.1	12	0	0.0060

Table 4 (continued)

Dolostone Ore Formation												
Stage 1						Stage 2						
	Median	Min	Max	N	n	DL	Median	Min	Max	N	n	DL
V	0.10	0.037	0.52	13	0	0.030	0.027	BDL	0.25	49	22	0.025
Cr	0.29	BDL	0.76	34	14	0.23	0.67	BDL	0.97	56	47	0.30
Mn	21	9.9	136	14	0	0.19	9.2	0.79	319	40	0	0.18
Co	12,432	4031	30,380	34	0	0.053	3699	1.3	102,748	57	0	0.038
Ni	246	16	2054	32	0	0.21	181	BDL	5893	56	1	0.23
Cu	BDL	BDL	230	19	10	1.6	1.9	BDL	183	53	22	1.4
Zn	BDL	BDL	BDL	11	11	4.9	BDL	BDL	18	56	50	4.9
Ga	0.090	BDL	1.0	25	3	0.027	0.044	BDL	1.2	63	15	0.024
Ge	0.36	BDL	1.7	32	1	0.082	0.38	BDL	1.5	63	6	0.11
As	0.60	BDL	2.9	29	14	0.34	5.2	BDL	9917	61	16	0.47
Se	23	6.0	60	33	0	3.4	31	BDL	175	64	3	4.0
Mo	BDL	BDL	0.12	30	18	0.034	0.077	BDL	3.1	62	37	0.031
Ag	0.53	BDL	1.5	25	2	0.011	0.61	BDL	3.4	62	13	0.012
Cd	BDL	BDL	BDL	34	34	1.5	BDL	BDL	BDL	62	62	1.5
In	BDL	BDL	0.032	28	18	0.017	BDL	BDL	0.26	62	41	0.010
Sn	BDL	BDL	1.6	34	20	0.26	BDL	BDL	26	63	38	0.23
Sb	0.076	BDL	0.67	32	13	0.054	0.15	BDL	235	64	24	0.057
Te	0.25	BDL	1.8	23	4	0.10	0.15	BDL	2.6	47	22	0.12
Au	BDL	BDL	BDL	14	14	0.10	BDL	BDL	0.12	64	46	0.013
Hg	BDL	BDL	1.5	30	18	0.36	0.70	BDL	22	63	27	0.51
Tl	3.7	0.019	10.9	24	0	0.010	0.26	BDL	3.8	64	10	0.0090
Pb	25	2.4	117	27	0	0.068	18	BDL	4404	62	6	0.17
Bi	1.5	0.31	4.3	23	0	0.0076	0.28	BDL	29	62	9	0.0086

was only identified in the Kupferschiefer and the Dolostone Ore Formation. Iron, Cd, Tl, In, Co, Mn, Pb, and Ni (listed from more to less suitable) are the most prominent discriminators for sphalerite between the two deposits. Although most trace element contents in pyrite are highly variable, favorably high Gini values are shown for certain elements. Manganese, Ag, Tl, V, As, and Co are suitable discriminating elements between pyrite from the Katanga Copperbelt and the Dolostone Ore Formation (ESM 5, Fig. S15).

Random Forest analyses of the Dolostone Ore Formation and the Katanga Copperbelt by stage subdivision show that chalcopyrite from the two main ore stages of the Katanga Copperbelt are discriminated best by Ga, Se, In, Sb, V, and Pb (ESM 5, Fig. S16). Selenium and Sb are elevated in stage 1, and Ga, In, Pb, and V in stage 2. Selenium, Co, Ni, Ga, and Bi are suitable discriminator elements for chalcopyrite from the two main ore stages of the Dolostone Ore Formation deposit. Stage 1 Dolostone Ore Formation chalcopyrite is enriched in Co and Bi, whereas stage 2 is enriched in Se, Ni, and Ga. Cobalt, Cd, Se, Fe, Ge, and Ga are the best discriminator elements for sphalerite from the two main ore stages of the Dolostone Ore Formation deposit, where stage 1 sphalerite is enriched in Co and Fe, and stage 2 sphalerite, in Se, Cd, Ga, and Ge.

Discussion

Distinction among deposits using chalcopyrite chemistry

Random Forest analyses demonstrate that 1) Ga and Ag are the best discriminators between Kupferschiefer and the Katanga Copperbelt; 2) Ga, Ge, and Zn between the Katanga Copperbelt and Dolostone Ore Formation; and 3) Tl and In between the Kupferschiefer and the Dolostone Ore Formation (Fig. 4a). Gallium is the most prominent discriminator of the Luiswishi deposit of the Katanga Copperbelt, with In and Sn being secondary discriminators (Fig. 4a). The fact that Ag is high for Kupferschiefer chalcopyrite is consistent with the fact that the Kupferschiefer is one of the world's largest producers of Ag from Cu ore (USGS 2020). Chalcopyrite can host hundreds µg/g of Ag (George et al. 2018). Silver contents also distinguish chalcopyrite the Dolostone Ore Formation and Central African Copperbelt as well as chalcopyrite from the Polish Kupferschiefer and the Central African Copperbelt, whereas Ga is strongly associated with Katanga Copperbelt when compared to the other two deposits (Fig. 4a). Gallium and Ag contents in chalcopyrite were plotted to test the results of Random Forest analyses (Fig. 5). Gallium, which

compared to the other two deposits (Fig. 4a and ESM 5, Fig. S15) has the highest Gini value for the Katanga Copperbelt data, clearly discriminates the Katanga Copperbelt samples. Silver has the highest Gini value only between the Polish Kupferschiefer and the Katanga Copperbelt, and discriminates these two areas well (Fig. 5). The Gini value is lower for Ag between the Dolostone Ore Formation and Katanga Copperbelt (Fig. 4a; ESM 5, Fig. S15), meaning it is not as powerful as a discriminating element (Fig. 5).

Distinction between mineralization stages using chalcopyrite compositions

Using results from Random Forest analyses, which was carried out only for trace element data from the Katanga Copperbelt and Dolostone Ore Formation datasets due to the risk of overfitting the smaller datasets of the other districts (Fig. 6a–b), there are obvious differences in the composition of chalcopyrite between stage 1 and 2 sulfides (e.g., Fig. 1; Table 2). Gallium, In, V, and Pb are the best indicator elements to distinguish the first and second mineralization stages in the Katanga Copperbelt (Fig. 6a). Stage 1 chalcopyrite is defined by higher contents of Se and Sb. It is important to note that several elements (e.g., Mn, Co, Ga, Ag, and Sb; Fig. 1) have significant variance between samples within one ore stage. Stage 1 from the Dolostone Ore Formation deposit yielded the highest mean Co Gini value whereas stage 2 has the highest value for Se (Fig. 6b). Although omitted from the Random Forest analyses, the Zambian Copperbelt Ga–Ag contents in chalcopyrite show two distinct populations (Fig. 5), the first population plotting with Dolostone Ore Formation stage 1 chalcopyrite and the second with Katanga Copperbelt stage 1 chalcopyrite. A possible explanation for the two distinct populations may be that both the Zambian Copperbelt and Dolostone Ore Formation experienced higher grades of regional metamorphism (Mendelsohn 1961; Annels 1986; Richards et al. 1988; Bertrandsson Erlandsson et al. 2022). Hence, the Zambian Copperbelt data may reflect both a stage 1 and 2 population, whereas one may reflect more low temperature formation conditions (plotting with Katanga Copperbelt, and the other higher temperature metamorphic conditions (the Dolostone Ore Formation deposit).

Distinctions among deposits using sphalerite chemistry

High Cd and Tl and low In characterize sphalerite from the Polish Kupferschiefer (Fig. 4b), in contrast to Dolostone Ore Formation sphalerite that is remarkably Cd-, Tl-, and Mn-poor but elevated with respect to Fe and In (Table 3). Ga

contents are similar between the two deposits. Manganese, Fe, and In in sphalerite have been shown to be temperature-dependent by the Ga–Ge–In–Mn–Fe in sphalerite (GGIMFis) geothermometer developed by Frenzel et al. (2016). Manganese and Fe increase whereas Ga and Ge decrease with higher temperatures (Frenzel et al. 2016). When the GGIMFis is applied to Dolostone Ore Formation sphalerite, formation temperatures above 310 ± 50 °C are calculated (Bertrandsson Erlandsson et al. 2022). Sphalerite (stage 2) from the Polish Kupferschiefer yielded formation temperatures of 122–123 °C (Foltyn et al. 2022). This is close to the temperature maximum of 100–130 °C for the main-stage stratiform sediment-hosted mineralization (Speczik and Puttmann 1987; Bechtel et al. 1995) and 145–200 °C for later epigenetic veins (Strengel-Martinez et al. 1993). The estimated sphalerite formation temperatures are consistent with trace element signatures. For example, Dolostone Ore Formation sphalerite is elevated in Mn, Fe, and In whereas being depleted in Ge in comparison to the Polish Kupferschiefer sphalerite (Fig. 2).

Distinction between mineralization stages using sphalerite compositions

The two main stages of mineralization at the Dolostone Ore Formation deposit include the Co-rich stage 1 and stage 2 that is characterized by a Cu–Zn association but lacks substantial Co-phases (Bertrandsson Erlandsson et al. 2022). The Random Forest analyses document this distinction because Co is affiliated with both chalcopyrite and sphalerite from stage 1 (Fig. 6b–c). Although the Random Forest analysis did not indicate Ni as an effective discriminator element for Dolostone Ore Formation sphalerite, high Ni in sphalerite was indicated by Bertrandsson Erlandsson et al. (2022), along with high Ni in chalcopyrite (Fig. 6b). The low Cd contents in Stage 1 Dolostone Ore Formation sphalerite suggest that leaching of host rocks may have been an important process because host rock limestones have low average Cd contents (0.04 µg/g; Chon and Shimazaki 1986). Schwartz (2000) suggested that low Cd contents in sphalerite correlate with high total activity of reduced S, which furthermore points to interaction with the reduced S from the sulfides or microbial S associated with the host sedimentary rocks. Ore stage 1 Dolostone Ore Formation sphalerite has a median Co content of 8,600 µg/g, with minimum and maximum contents of 4,300 and 10,700 µg/g respectively (Table 3). The later ore stage 2 Dolostone Ore Formation sphalerite, associated with late orogenic remobilization, has a much wider range of Co contents (from 32 to 10,100 µg/g). It is for this reason the Random Forest analyses does not attribute Co a higher mean Gini value. Nonetheless, Cd

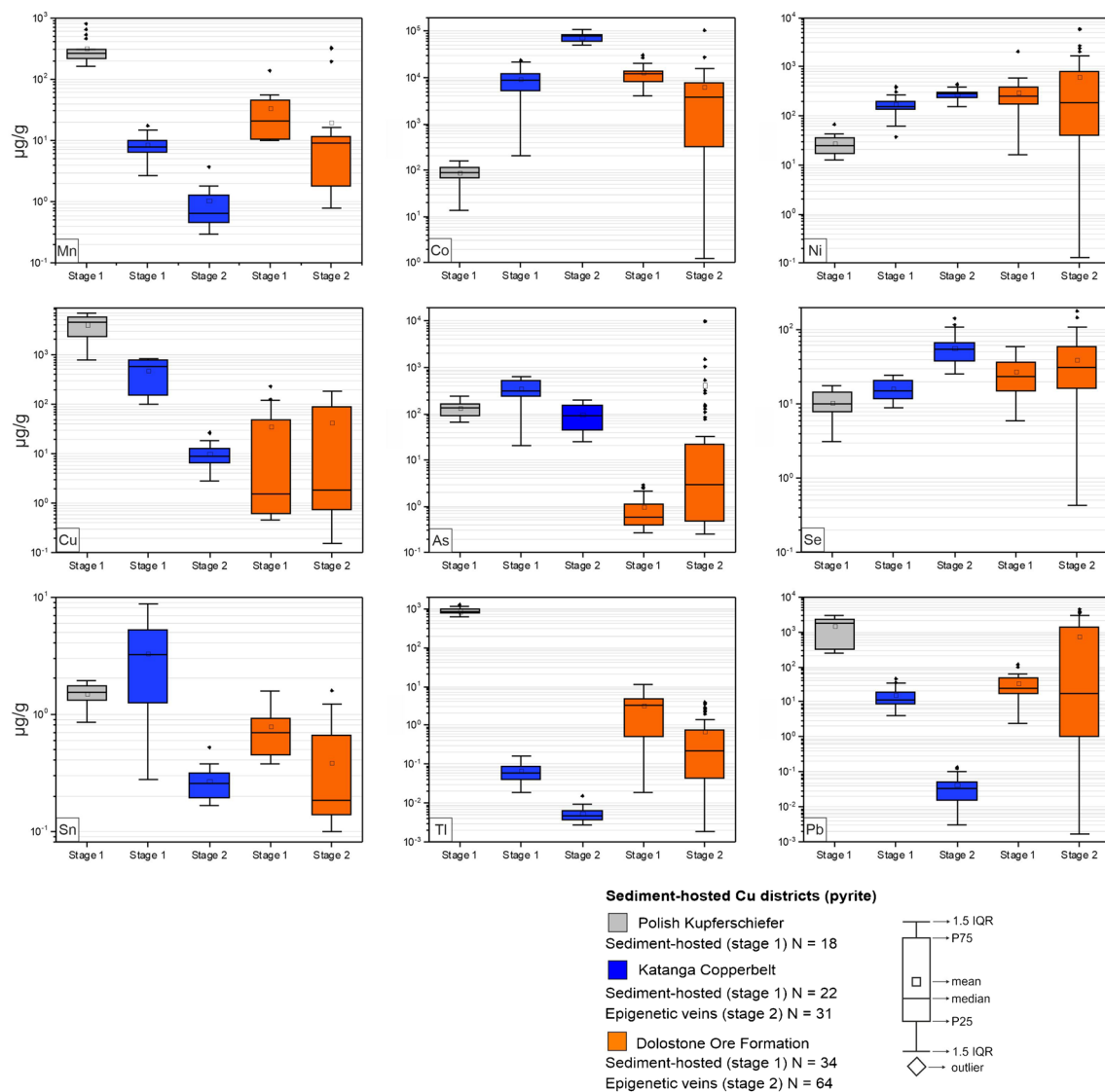


Fig. 3 Pyrite trace element geochemistry of selected elements from the Polish Kupferschiefer, Katanga Copperbelt, and the Dolostone Ore Formation. Data for the Dolostone Ore Formation is from this

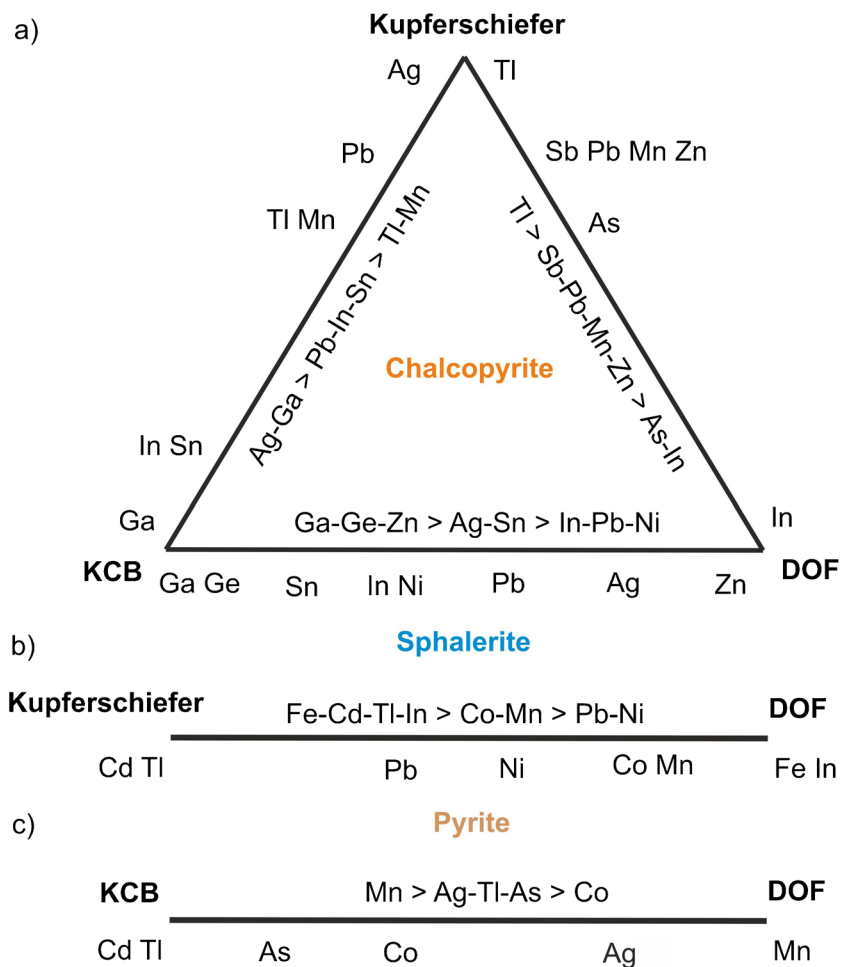
study and Bertrandsson Erlandsson et al. (2022). N=number of LA-ICP-MS analyses for each group

and Co are interesting elements for discriminating both the stages and deposits, as these are not believed to be directly temperature dependent. A noteworthy point is that Co and Cd were included in the study by Frenzel et al. (2016), but they could not explain the controlling process(es) of these elements in sphalerite. However, Co and Cd are suitable discriminating elements between the Polish Kupferschiefer and Dolostone Ore Formation. The Random Forest analyses indicated Fe and Cd to be the two best discrimination elements (Fig. 7). From this, it is also evident that stage 2 sphalerite is characterized by lower Fe and higher Cd, compared to their stage 1 equivalent.

Distinctions between deposits using pyrite chemistry

Manganese, Cd, and Tl are the best discriminator element for pyrite between the Katanga Copperbelt and the Dolostone Ore Formation (Fig. 4c) Pyrites from both deposits show high contents of Co (Fig. 3; Table 4), although the Katanga Copperbelt has the highest median Co contents of 5.6 wt% compared to the median 6,445 µg/g Co in Dolostone Ore Formation pyrite and 91 µg/g in the Polish Kupferschiefer. It is obvious that pyrite from the Co-rich districts is significantly rich in Co. This is also reflected in stage

Fig. 4 Binary and ternary diagrams of deposit-defining elements, derived from Random Forest analyses, of (a) chalcopyrite, (b) sphalerite, and (c) pyrite. The proximity of the element to the deposit name relates to a higher mean Gini value. Relative grouping of the elements, based on higher Gini values, are indicated by >. Abbreviations: KCB = Katanga Copperbelt; DOF = Dolostone Ore Formation



2 Dolostone Ore Formation pyrite, which contains only median 3,699 µg/g Co, whereas Dolostone Ore Formation stage 1 pyrite has median 12,432 µg/g Co. Stage 2 Dolostone Ore Formation pyrite lacks associated Co-mineralization (Bertrandsson Erlandsson et al. 2022). Pyrite from Katanga Copperbelt stage 1 also formed before the formation of stoichiometric Co-minerals (e.g., Cailteux et al. 2005), which would explain the lower Co contents in Katanga Copperbelt stage 1 pyrite (8,480 µg/g) compared to stage 2 (7.6 wt%). Pyrite from these deposits is believed to have formed before the main Co-mineralization, indicating that the Co-enrichment in pyrite occurred throughout the multiple stages of pyrite formation. Similar trends appear to be true for Cu. Copper is elevated in pyrite from both the Polish Kupferschiefer (median 4,593 µg/g) and the stage 1 Katanga Copperbelt (median 564 µg/g); both these pyrite stages predate first Cu mineralization. In contrast, stage 2 Katanga Copperbelt contains median 10.3 µg/g and both Dolostone Ore Formation stages have Cu contents below or around detection limits (4.9 µg/g). The stage 1 ore-forming fluids must hence have been elevated in Cu during pyrite formation. Although

obvious inclusions were avoided in the data processing, the significantly elevated contents of Cu and Ag in pyrite from the Polish Kupferschiefer may also be attributed to chalcopyrite (e.g., nano-) inclusions, as the chalcopyrite is Ag-rich (Foltyn et al. 2022). Hence, pyrite chemistry tends to mirror the associated metals of the mineralization, where the Polish Kupferschiefer pyrites are enriched in, for example, Ag whereas the African Co-deposits have pyrite with elevated Co contents. Pyrite may thus serve as an important exploration tool and indicator mineral for stratiform sediment-hosted Cu and Co ores.

Comparison of trace elements compositions to other deposit types

Element groups within different sulfide minerals are similar between the investigated deposits. For example, chalcopyrite from all deposits has an association of Co, Ni, Se, and Bi, sphalerite is characterized by Ga, Ge, Se, and Hg, and pyrite by Ni and Se. Thus, trace element compositions may serve

Fig. 5 Binary plot of Ga and Ag contents in chalcopyrite from the four stratiform sediment-hosted Cu(-Co) districts investigated in this study. Gallium and Ag were used because they proved to be good discriminator elements by Random Forest analyses

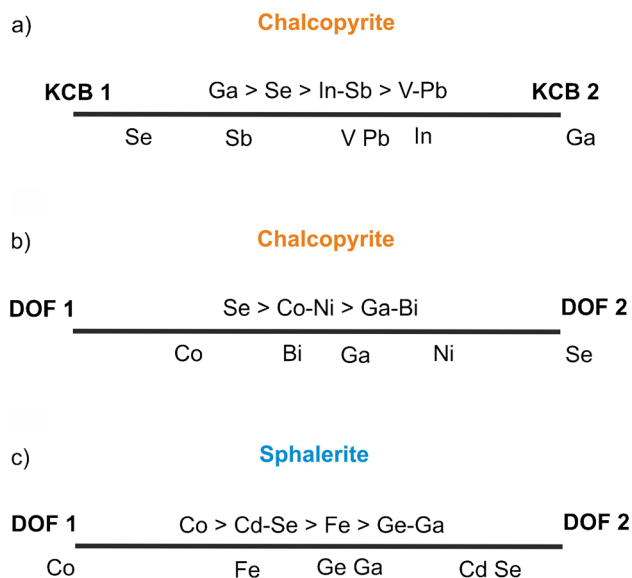
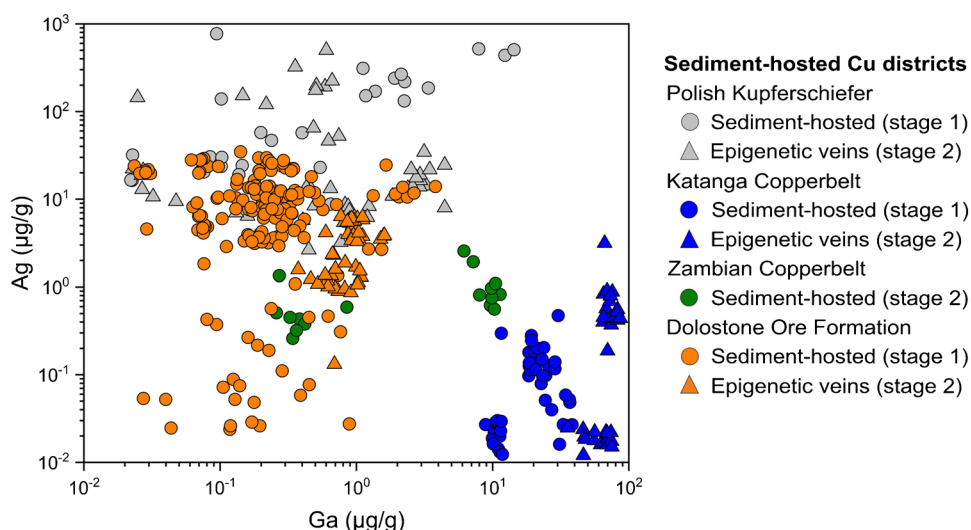


Fig. 6 Binary representation of defining elements, from Random Forest analysis, in chalcopyrite (**a** and **b**) between the two main mineralization stages of the Katanga Copperbelt (KCB 1 and 2) and Dolostone Ore Formation (DOF 1 and 2). And in sphalerite (**c**) between the two main mineralization stages of the Dolostone Ore Formation deposit. Ore stage 1 (DOF 1) being the main Cu-Co-Zn stage and ore stage 2 (DOF 2) being Cu-Zn mineralization without substantial Co mineralization. The proximity of the element, underneath the thick line, to the deposit name relates to a higher mean Gini value. Relative grouping of the elements, based on higher Gini values, are indicated by >

to characterize stratiform sediment-hosted Cu(-Co) deposits, as these trends comprise both early and late stages of mineralization. Discriminating plots are available for pyrite and sphalerite from different ore deposit types (magmatic, hydrothermal, sedimentary, MVT, VHMS, Sedex, Skarn), which allows a comparative study. The limited discrimination plots

that exist for chalcopyrite (Duran et al. 2019) are based on elements that are below respective detection limits for the vast majority of the collected data (e.g., Zn and Cd).

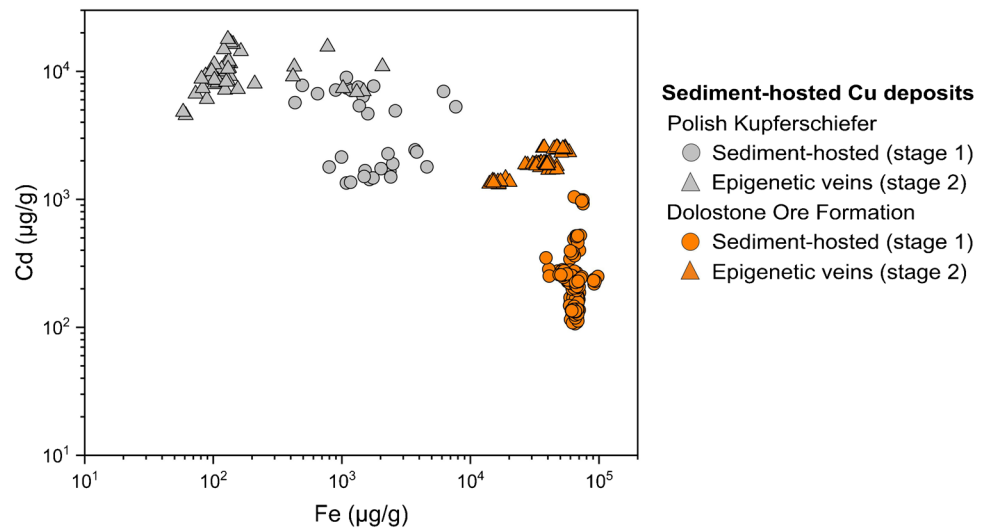
Sphalerite as a discriminator of deposit types

Liu et al. (2022) compiled sphalerite trace element data to develop discrimination plots for MVT, VHMS, skarn, and SEDEX sphalerite. Both stage 1 Kupferschiefer and stage 2 Dolostone Ore Formation sphalerite plot predominantly within the MVT field based on Ge versus Mn, and Fe versus Mn (Fig. 8). However, some stage 1 Dolostone Ore Formation sphalerite compositions fall in the higher-temperature skarn or VMS fields (Fig. 8). This is consistent with the fact that Mn, Fe, and Ge are all temperature dependent with high contents of Mn and Fe attributed to higher formation temperatures (Frenzel et al. 2016). The Dolostone Ore Formation stage 1 sphalerite that trends towards the higher-temperature fields (Fig. 8) as it formed at temperatures $> 310 \pm 50$ °C (Bertrandsson Erlandsson et al. 2022).

Pyrite as a discriminator of deposit type

Cobalt-Ni ratios are among the most utilized trace element pairs to determine the origin of pyrite from the early work of Loftus-Hills and Solomon (1967) discriminating sedimentary, volcanic (i.e., hydrothermal), and magmatic pyrite. Campbell and Ethier (1984) proposed a Co-Ni discrimination plot with fields for pyrite of sedimentary, volcanic/hydrothermal, and magmatic origin. The stratiform sediment-hosted Cu-Co pyrite of this study clusters together in a separate field (Fig. 9a) because all pyrites analyzed have high Co and Ni contents that are lower than the magmatic pyrite field. Therefore we propose an additional field for pyrite of sedimentary Cu-Co affinity.

Fig. 7 Binary plot of Fe and Cd contents in sphalerite from the Polish Kupferschiefer and the Namibian Dolostone Ore Formation



Duran et al. (2015) used Co, As, Se, and Sb trace element data to distinguish between hydrothermal deposits (Cu-porphyrries, orogenic Au, and volcanogenic-hosted massive sulfides) and magmatic deposits. Due to high Co values, our stratiform sediment-hosted Cu-Co pyrite data plot within the magmatic field of this discrimination plot (Fig. 9b). Sedimentary pyrite from both the Dolostone Ore Formation and the average Neoproterozoic sedimentary pyrite from Gregory et al. (2015) plot in a gap between the hydrothermal and magmatic fields, perhaps reflecting sedimentary to diagenetic pyrite versus hydrothermal and magmatic pyrite. The source of Co in stratiform sediment-hosted Cu-Co deposits is still enigmatic and several sources have been suggested: mafic volcanoclastics (Unrug 1988), erosion materials of basement-hosted magmatic Cu-Co–Ni ores (Cailteux et al. 2005), or the basement rocks in general (El Desouky et al. 2010; Van Wilderode et al. 2015). These sources could thus explain the magmatic trace element signatures seen in pyrite, as other magmatic affiliated elements might also have been transported in the metal-scavenging fluids, resulting in an inherited magmatic trace element signature in the stratiform sediment-hosted Cu-Co pyrite.

Thallium is significantly higher in the Polish Kupferschiefer pyrite (846 μg/g) compared to the other districts (Fig. 3). Interestingly, pyrite from SEDEX deposits in the McArthur Basin in Australia is characterized by Tl enrichment (Mukherjee and Large 2017), and the Tl content in pyrite from Kupferschiefer is comparable to those reported from the Teena deposit (Magnall et al. 2022). Polish Kupferschiefer pyrite also contains elevated contents of Sb and Hg. High contents of As, Hg, Sb, and Tl are common in pyrite from Alpine-type, low-temperature, carbonate-hosted Pb–Zn deposits in the Eastern Alps (Melcher et al. 2023). Several sediment-hosted Carlin-type Au districts also show a positive As–Sb–Hg–Tl trend (e.g., Muntean et al. 2011;

Daliran et al. 2018). Another example is the sediment-hosted As–Sb–Tl–Pb ± Hg ± Au occurrence of Janjevo, Kosovo that is attributed to a distant, unknown porphyry system (Mederski et al. 2022). The source of these Carlin-like As–Sb–Tl–Pb ± Au mineralizing fluids was suggested by several authors to be fueled by subduction-related magmatism (e.g., Muntean et al. 2011; Mederski et al. 2022). This again suggests that pyrite chemistry inherits trends related to the metal-source, without direct proximity to, for example, a magma chamber.

Impact of metal sources on sulfide trace element composition in stratiform sediment-hosted Cu(-Co) deposits

Trace element contents in sulfides of hydrothermal origin depend on many factors that can be condensed to three essential aspects. The first are parameters of the fluids such as temperature, pressure, pH, salinity, and oxidation state that control how much of a given element can be transported. The second is the source of metals, defining how much of a given element is available for transport. The third is related to the specifics of mineral deposition, including both the type of process (e.g., boiling, cooling, fluid mixing, redox change etc.) and co-crystallizing phases that can affect element partitioning. Later processes, such as metamorphism, could result in remobilization of certain elements. Although this study has treated the investigated deposits as one deposit type (stratiform sediment-hosted Cu), it is worth keeping in mind that there are significant differences between European and African deposits. Kupferschiefer deposits in Poland and Germany are Cu–Ag rich and relatively undeformed, whereas the African deposits are distinctively Cu–Co rich and underwent significant deformation. Diagenetic mineralization in

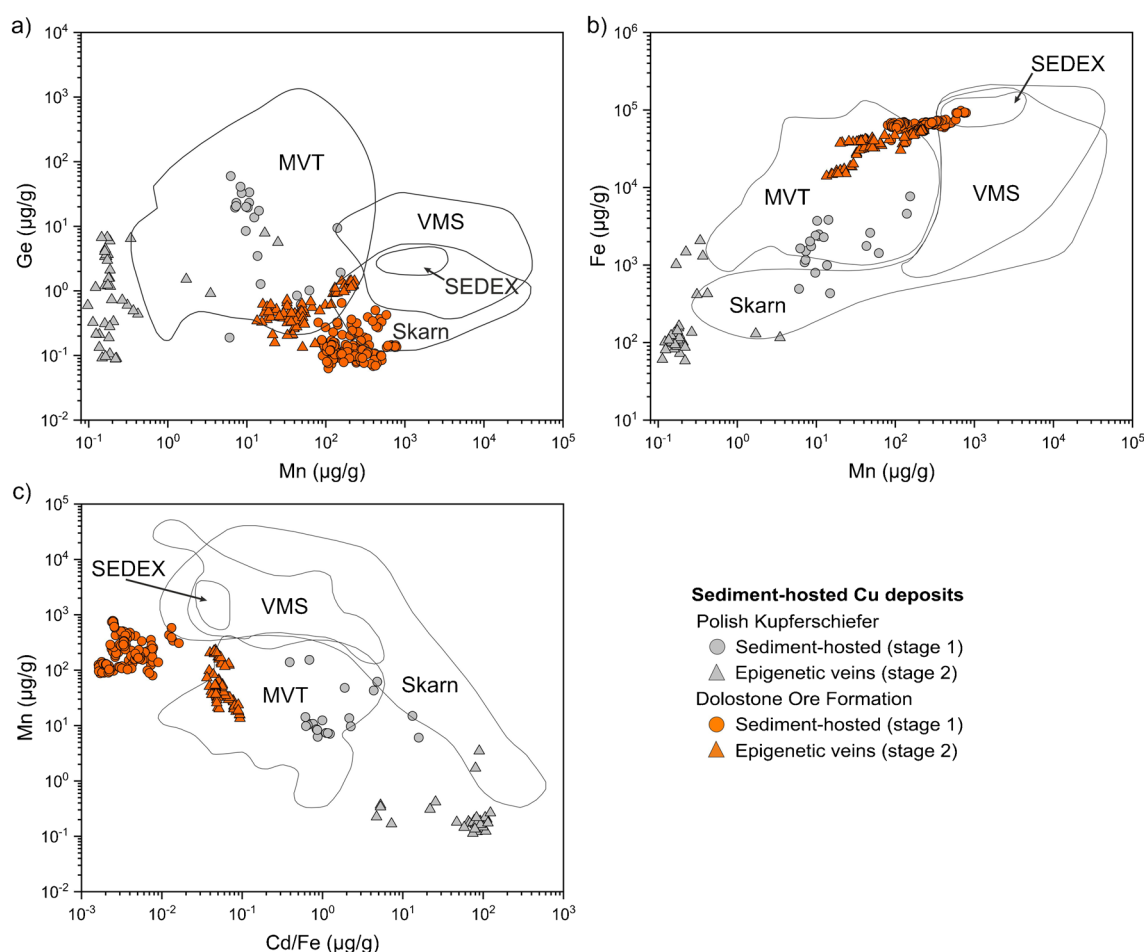


Fig. 8 Binary discrimination plots of (a) Mn vs Ge, (b) Mn vs Fe, and (c) Cd/Fe vs Mn for sphalerite from MVT deposits and higher temperature deposits (VMS, skarn, and SEDEX), fields after Liu et al. (2022) and compiled reference data therein

both districts likely formed under similar temperature, pH, and Eh conditions, but the Central African Copperbelt also experienced syn- and post-tectonic mineralizing events. This resulted in the replacement of earlier sulfides and formation of new sulfides at significantly higher temperatures (e.g., El Desouky et al. 2009), evident by elevated In contents in chalcopyrite and Fe and In in sphalerite from the African districts (Fig. 4a). The case studies of the stratiform sediment-hosted Cu(-Co) deposits in this investigation were deliberately chosen to enable preliminary assessment of varying processes and sources. Katanga Copperbelt samples allow assessment of differences between stage 1 and 2 fluids, whereas comparison of Kupferschiefer and Katanga Copperbelt stage 1 provides clues regarding how diagenetic mineralization is affected by metamorphism.

Comparison of trace elements in chalcopyrite, sphalerite, and pyrite suggests that initial high Tl and Sb (and potentially Ag) contents exemplified by Kupferschiefer samples, could be remobilized during anchimetamorphism (case of Katanga Copperbelt stage 1). Metamorphic-orogenic related

deposits (Katanga Copperbelt stage 2, Zambian Copperbelt, Dolostone Ore Formation) exhibit much lower contents of these elements. For example, pyrite from the particularly well preserved (essentially unmetamorphosed) Teena SEDEX deposit shows enrichment in Tl, Sb, Ag in addition to Cu, Zn, As, and Pb, compared with pyrite from the metamorphosed McArthur basin (Magnall et al. 2022). Indeed, the data show more similarities between the African deposits compared to the Polish Kupferschiefer, and this is very likely due to the difference in metal sources for the metallogenic districts investigated herein, especially for elements such as In, Sn, Ag, and Co.

Metal sources for the Polish Kupferschiefer

The extraordinary size of the Kupferschiefer Cu-Ag district requires significant metal sources. Wedepohl et al. (1978) investigated Pb isotopes that indicate basement rocks as a major source of Pb in the Kupferschiefer; however, this was

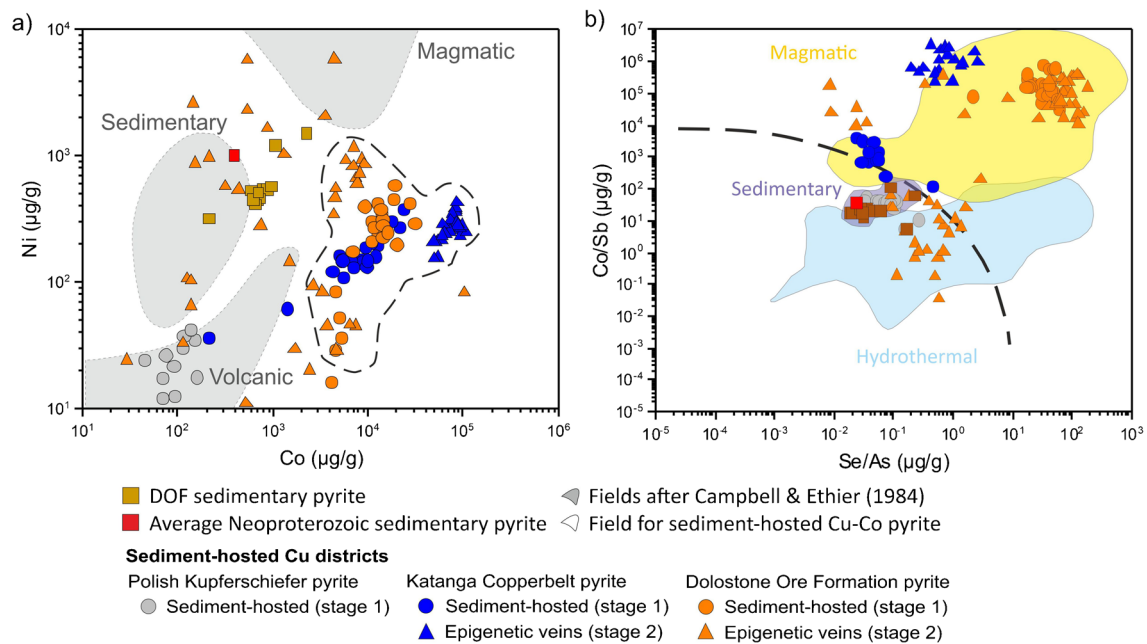


Fig. 9 a) Binary discrimination plot of sedimentary, volcanic (i.e., hydrothermal), and magmatic pyrite based on Co–Ni contents in pyrite (after Campbell and Ethier 1984). A new field for stratiform sediment-hosted Cu–Co pyrite is herein proposed. Sedimentary Dolostone Ore Formation pyrite is shown separate along with a data point for average Neoproterozoic sedimentary pyrite from Gregory et al. (2015) as reference points in the plot. Dolostone Ore Formation pyrite data includes combined data from this study and Bertrandsson Erlandsson et al. (2022). b) Binary discrimination plot of

hydrothermal and magmatic pyrite based of Se/As and Co/Sb (after Duran et al. 2015). Sedimentary Dolostone Ore Formation pyrite is shown separate along with a data point for average Neoproterozoic sedimentary pyrite from Gregory et al. (2015) as reference points for sedimentary pyrite field suggested in the study. Dolostone Ore Formation pyrite data includes combined data from this study and Bertrandsson Erlandsson et al. (2022). Abbreviations: DOF=Dolostone Ore Formation

interpreted in the light of the older syngenetic origin model, where eroded country rocks provide metals transported to the Zechstein Sea by rivers. More recent diagenetic-epigenetic models conclude that Cu and other metals were liberated by highly saline brines from the volcano-sedimentary sequence filling lower Permian basins (the Upper Rotliegendes red beds and bimodal volcanic units in the Lower Rotliegendes). These models, however, do not exclude the possibility that underlying basement rocks, which are accessible by faults and shear zones, might also have played a role as metal sources (Jowett 1986; Oszczepalski 1989, 1999; Borg 1991; Cathles et al. 1993; Blundell et al. 2003; Hitzman et al. 2005; Brown 2009; Borg et al. 2012).

If the elemental signature of these sources is distinct enough, the trace elements study of ore minerals might give insight to the contribution from Permian rocks and the basement. In this regard the Myszków Mo–Cu–W deposit in Poland is worth mentioning, a rare example of Variscan-related porphyry-style mineralization, which formed synchronous with felsic magmatism at 303.8 – 292.7 Ma (Mikulski and Stein 2012; Mikulski et al. 2019), located approximately 220 km SE from Lubin. It was partly eroded and then covered by a sequence of Triassic marine rocks

(Podemski 2001), meaning it likely contributed to the sedimentation of the Rotliegend strata. Rhenium is an enigmatic commodity of the Polish Kupferschiefer that has been shown to be hosted in chalcopyrite and djurelite by LA-ICP-MS (Foltyn et al. 2022). Porphyry Cu–Mo deposits are a major source of Re (U.S. Geological Survey 2020); thus, leaching metals from the sediments that formed from erosion and redeposition of such deposits may partially preserve element signatures of those deposit types. Additionally, Ag tellurides and Ag, Pb sulfosalts have been described from veinlet-impregnation mineralization around Variscan granitoid intrusions associated with the Myszków porphyry deposit (Mikulski et al. 2019). Such deposits would be a reasonable, source of, for example, Ag and Pb, which are elevated in the Kupferschiefer chalcopyrite, sphalerite, and pyrite compared to the other districts (Figs. 1, 2, 3, 4a; Table 4).

Metal sources for the African stratiform sediment-hosted Cu–Co deposits

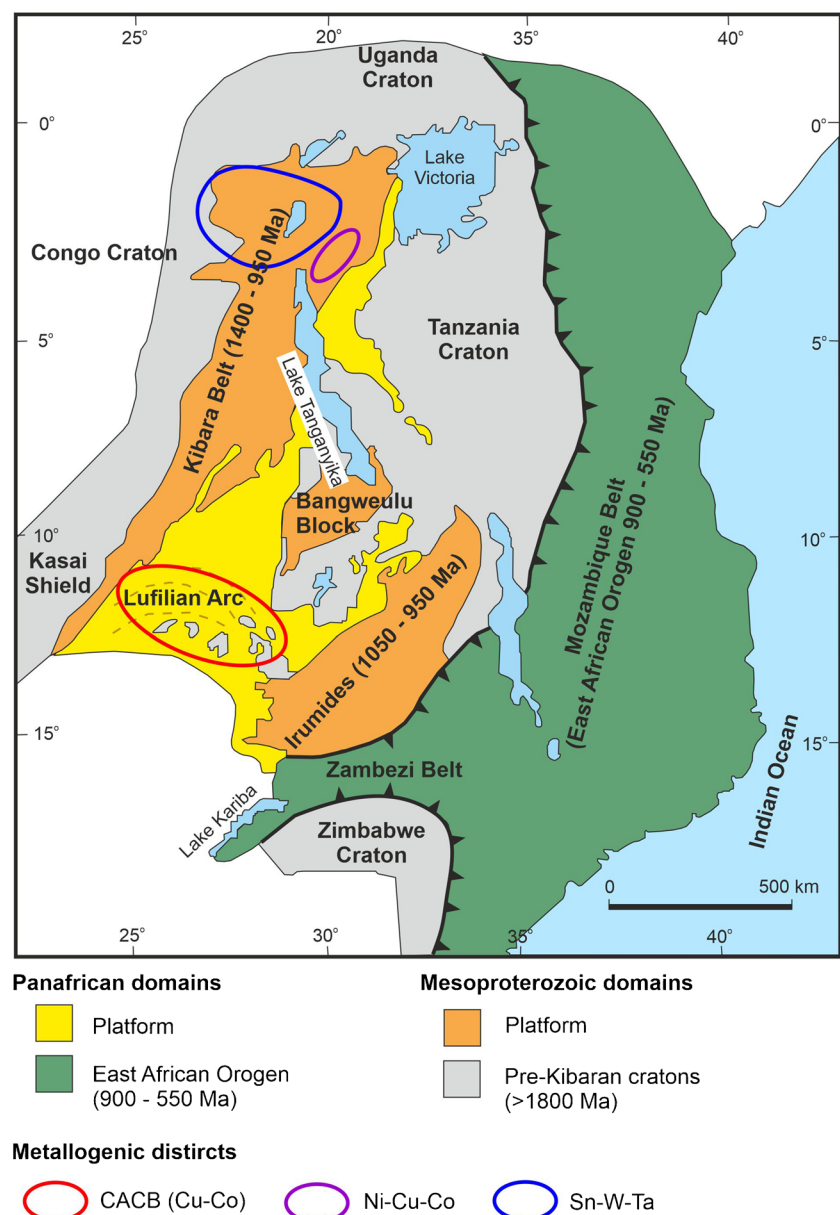
In contrast to the Kupferschiefer, the Central African Copperbelt and the Dolostone Ore Formation have a longer and more complex geological history with multiple orogenic

and metamorphic events. Therefore, the sources for the metals may vary. Whereas the source for the Cu-Co-Zn in the Dolostone Ore Formation is highly uncertain, it is widely believed that Cu-Co in the Central African Copperbelt is predominantly sourced from mafic volcanic and basement rocks or from the eroded products of basement rocks (e.g., Selley et al. 2005; Van Wilderode et al. 2014). As mafic rocks have higher contents of elements such as Co and Cu (Faure 1997), this is a more plausible metal source than for the Polish Kupferschiefer.

As the basement in the larger geographical area around the Central African Copperbelt contains Cu-Co-Ni ores hosted by layered mafic-ultramafic intrusion, these deposits (and their eroded detrital sediments) have been proposed

to be a possible metal source for stratiform sediment-hosted Cu-Co deposits in the Central African Copperbelt (Cailteux et al. 2005; Kampunzu et al. 2009). Musongati and Kabanga are two examples of Ni-Cu-Co deposits situated in the Kibara Belt, north of the Central African Copperbelt (Pohl et al. 2013; Fig. 10). The Kibara Belt is a Mesoproterozoic NE-SW trending orogenic belt that stretches from the west of Lake Victoria southward and underneath the Lufilian Arc, where the Central African Copperbelt is situated (Fig. 10). The elevated contents of Sn in chalcopyrite from the Central African Copperbelt (Fig. 1) support this model, as there is a vast amount of post-orogenic granite-related Sn(-W-Ta) occurrences also present in the Kibara Belt (Pohl et al. 2013; Hulsbosch et al. 2014; Fig. 10). The possible occurrence of

Fig. 10 Metallogenic districts containing elements associated with the sulfide trace element compositions in the stratiform sediment-hosted copper deposits of the Central African Copperbelt (CACB). Geological units simplified after De Waele et al. (2008), metallogenic districts after Pohl et al. (2013)



orogenic Sn-granites (Caron et al. 1986) in the basement could explain the Sn enrichment in Central African Copperbelt chalcopyrite (Fig. 1).

Chalcopyrite from both the Katanga and Zambian copperbelts shows elevated contents of Ga and chalcopyrite from the Katanga Copperbelt shows elevated Ge compared to the other districts (Fig. 1). The stratiform sediment-hosted Cu(-Co) districts are not the only example of ore deposits with elevated Ga and Ge contents: the carbonate-hosted Kipushi deposit in Katanga and the Tsumeb deposit in Namibia are well known for the occurrence of Ga- and Ge-phases and high contents of these elements in chalcopyrite and sphalerite (Kelvin et al. 2022). In particular within the Katanga part, it is evident that the Central African Copperbelt is rich in Ga and Ge occurrences, which is seen in the sulfides of the stratiform sediment-hosted Cu-Co deposits, although the Kipushi deposit formed much later than the stratiform sediment-hosted Cu-Co deposits (Schneider et al. 2007; Heijlen et al. 2008). Van Wilderode et al. (2013) suggested that the sources of Ga and Ge in the Kipushi deposit were both mafic and felsic basement rocks, and possibly siliciclastic sedimentary rocks of the Roan Group. Hence, a similar source for Ga and Ge in the stratiform sediment-hosted Cu-Co districts seems plausible.

Conclusions

Random Forest analysis of chalcopyrite attributes Ag and Tl to the Polish Kupferschiefer, Ga and Ge to the Katanga Copperbelt and Zn and In to the Dolostone Ore Formation deposit as top deposit indicator elements. Cobalt, Ni, and Se in chalcopyrite show similar contents between the four investigated districts. The mean contents of Ga in chalcopyrite ($< 10 \mu\text{g/g}$) are low in all deposits except the Katanga Copperbelt (median of $32 \mu\text{g/g}$). Arsenic, Ag, In, Sn, and Sb show potential in discriminating chalcopyrite from stratiform sediment-hosted Cu deposits from copper-rich sediment-hosted deposits with high Co.

Cadmium, Tl, Fe, and In contents in sphalerite characterize the Polish Kupferschiefer and the Dolostone Ore Formation deposit based on Random Forest analysis. Both Fe and In are strongly temperature dependent in sphalerite (Frenzel et al. 2016) which helps to explain this difference. Sphalerite from the Polish Kupferschiefer shows the lowest average contents of Mn, Fe, and In compared to other deposit types. Trace element contents in sphalerite also illustrate distinctions between stages. Cobalt is only highly enriched in Co-rich stage 1 sphalerite from the Dolostone Ore Formation deposit. Random Forest analysis of pyrite attributes As and Mn to the Katanga Copperbelt and Dolostone Ore Formation deposit, respectively. Cobalt is highly enriched in pyrites and shows Co/Ni ratios distinguishable from that of volcanic (i.e., hydrothermal), magmatic, and sedimentary pyrite.

Our data also indicate that the mineral chemistry of sulfides from sediment-hosted Cu(-Co) deposits may be heavily impacted by different metal sources in the host basins. Porphyry Cu style-related deposits (including intrusion related veins) in the basement of the Kupferschiefer can explain higher contents of Ag and Pb in the sulfides. The presence of layered mafic-ultramafic Cu-Co-Ni and post-orogenic granite-related Sn(-W-Ta) deposits in the basement of the Central African Copperbelt is reflected in higher contents of Ni and Sn in the chalcopyrite, and Co in pyrite. The African districts (Dolostone Ore Formation deposit and Copperbelt) are situated in similar tectonostratigraphic settings and both were subjected to deformation and metamorphic processes, and therefore it is not surprising that these districts have more similarities to each other than to the Polish Kupferschiefer.

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Declarations

Conflict of Interest The authors declare no competing interests.

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References

- Alderton DHM, Selby D, Kucha H, Blundell DJ (2016) A multi-stage origin for Kupferschiefer mineralization. *Ore Geol Rev* 79:535–543
- Allen NK (2016) Structurally controlled Cu-Zn-(Co-Pb) mineralization in the Neoproterozoic Ombombo Subgroup, Kaokoland, Namibia. Master Thesis. Colorado School of Mines
- Annels AE (1986) Ore genesis in the Zambia copperbelt, with particular reference to the northern sector of the Chambishi basin. *Can Mineral: Mineral Assoc Can. Cityview* 78087. Canada
- Armstrong RA, Master S, Robb LJ (2005) Geochronology of the Nchanga Granite, and constraints on the maximum age of the Katanga Supergroup, Zambian Copperbelt. *J Afr Earth Sci* 42:32–40
- Barra F, Broughton D, Ruiz J, Hitzman M (2004) Multi-stage mineralization in the Zambian Copperbelt based on Re-Os isotope constraints. *Geol Soc Am Abstr Progr* 36(5):516
- Bartholomé P, Evrard P, Katekesha F, Lopez-Ruiz J, Ngongo M (1972) Diagenetic ore-forming processes at Kamoto, Katanga, Republic of the Congo. Amstutz GC, Bernard AJ. *Ore sediments*. Springer-Verlag, New York, pp 21–41
- Bartholomé P, Katekesha F, Ruiz J, López, (1971) Cobalt zoning in microscopic pyrite from Kamoto, Republic of the Congo (Kinshasa). *Miner Deposita* 6(3):167–176
- Bechtel A, Elliott WC, Wampler JM, Oszczepalski S (1999) Clay mineralogy, crystallinity, and K-Ar ages of illites within the Polish Zechstein Basin; implications for the age of Kupferschiefer mineralization. *Econ Geol* 94(2):261–272
- Bechtel A, Püttmann W, Hoernes S (1995) Reconstruction of the thermal history of the Kupferschiefer within the Zechstein basin of central Europe: a stable isotope and organic geochemical approach. *Ore Geol Rev* 9(5):371–389
- Belissant R, Boiron MC, Luais B, Cathelineau M (2014) LA-ICP-MS analyses of minor and trace elements and bulk Ge isotopes in zoned Ge-rich sphalerites from the Noailhac–Saint-Salvy deposit (France): insights into incorporation mechanisms and ore deposition processes. *Geochim Cosmochim Acta* 126:518–540
- Benites D, Torró L, Vallance J, Laurent O, Valverde PE, Kouzmanov K, Chelle-Michou C, Fontboté L (2021) Distribution of indium, germanium, gallium and other minor and trace elements in polymetallic ores from a porphyry system: the Morococha District, Peru *Ore Geol Rev* 136:104236
- Bertrandsson Erlandsson V, Wallner D, Ellmies R, Raith JG, Melcher F (2022) Trace element composition of base metal sulfides from the sediment-hosted Dolostone Ore Formation (DOF) Cu-Co deposit in northwestern Namibia: Implications for ore genesis. *J Geochem Explor*, 107105
- Bieńko T, Pietrzela A (2022) Trace element distribution and geochemical zonation in the world-class Nowa Sól sediment-hosted Cu-Ag deposit, SW Poland. *Miner Deposita* 57:827–851
- Blundell DJ, Karnkowski PH, Alderton DHM, Oszczepalski S, Kucha H (2003) Copper mineralization of the Polish Kupferschiefer: a proposed basement fault-fracture system of fluid flow. *Econ Geol* 98(7):1487–1495
- Borg G (1991) The significance of Rotliegend volcanics for the metal provinces of the Kupferschiefer Basin. *Zbl Geol Paläont Teil I*, H 4:929–943
- Borg G, Piestrzyński AB, Gerhard H, Püttmann W, Walther S, Fiedler M (2012) An overview of the European Kupferschiefer deposits. In *Geology and Genesis of Major Copper Deposits and Districts of the World Economic Geology Special Publication*. Francisco Camus: 455–486
- Breiman L (2001) Random Forests. *Mach Learn* 45(1):5–32
- Brems D, Muchez P, Sikazwe O, Mukumba W (2009) Metallogenesis of the Nkana copper–cobalt south orebody, Zambia. *J Afr Earth Sci* 55:185–196
- Brown AC (2009) A process-based approach to estimating the copper derived from red beds in the sediment-hosted stratiform copper deposit model. *Econ Geol* 104(6):857–868
- Cailteux JLH, Kampunzu AB, Lerouge C, Kaputo AK, Milesi JP (2005) Genesis of sediment-hosted stratiform copper–cobalt deposits, central African Copperbelt. *J Afr Earth Sci* 42:134–158
- Cailteux JLH, Muchez P, Cuyper J, Dewaele S, Putter T (2018) Origin of the megabreccias in the Katanga Copperbelt (DR Congo). *J Afr Earth Sci* 140:76–93
- Campbell FA, Ethier VG (1984) Nickel and cobalt in pyrrhotite and pyrite from the Faro and Sullivan orebodies. *Can Mineral* 22(3):503–506
- Caraballo E, Beaudoin G, Dare S, Genna D, Petersen S, Relvas JMRS, Piercey SJ (2023) Trace element composition of chalcopyrite from volcanogenic massive sulfide deposits: variation and implications for provenance recognition. *Econ Geol* 118(8):1923–1958
- Caron JPH, Kampunzu AB, Lwango LB, Manteka B, Nkanika WR (1986) Les ressources minérales d'âge protérozoïque moyen en Afrique équatoriale et l'évolution géodynamique de la chaîne Kibarienne. UNESCO. *Geol Serv Dév Econ Bull* 5:139–152
- Cathles LM, Oszczepalski S, Jowett EC (1993) Mass balance evaluation of the late diagenetic hypothesis for Kupferschiefer Cu mineralization in the Lubin Basin of southwestern Poland. *Econ Geol* 88(4):948–956
- Celsius Resources Limited (2021) ASX Release 1 July 2021 Celsius doubles mineral resource at Opuwo cobalt-copper project. Internal report from: [celsiusresources.com](https://www.celsiusresources.com)
- Chen F, Deng J, Wang Q, Huizenga JM, Li G, Gu Y (2020) LA-ICP-MS trace element analysis of magnetite and pyrite from the Hetaoping Fe-Zn-Pb skarn deposit in Baoshan block, SW China: Implications for ore-forming processes. *Ore Geol Rev* 117:103309
- Chon HT, Shimazaki H (1986) Iron, manganese and cadmium contents of sphalerites and their genetical implications to hydrothermal metallic ore deposits in Korea. *Econ Environ Geol* 19:139–149
- Cook NJ, Ciobanu CL, George LL, Zhu ZY, Wade B, Ehrig K (2016) Trace element analysis of minerals in magmatic-hydrothermal ores by laser ablation inductively-coupled plasma mass spectrometry: approaches and opportunities. *Minerals* 6:111
- Cook NJ, Ciobanu CL, Pring A, Skinner W, Shimizu M, Danyushevsky LV et al (2009) Trace and minor elements in sphalerite: a LA-ICPMS study. *Geochim Cosmochim Acta* 73(16):4761–4791
- Daliran F, Hofstra A, Walther J, Topa D (2018) Ore genesis constraints on the Agdarreh and Zarshouran Carlin-style gold deposits in the Takab region of northwestern Iran. Chapter 9 in: *Diversity in Carlin-Style Gold Deposits*. *Rev Econ Geol*
- Daly MC, Chakraborty SK, Kasolo P, Musiwa M, Mumba P, Naidu B et al (1984) The Lufilian arc and Irumide belt of Zambia: results of a geotraverse across their intersection. *J Afr Earth Sci* 2(4):311–318
- Development Core Team R (2013) R: A language and environment for statistical computing. Vienna, Austria. ISBN 3–900051–07–0: URL: <http://www.R-project.org/>
- Dewaele S, Muchez P, Vets J, Fernandez-Alonzo M, Tack L (2006) Multiphase origin of the Cu–Co ore deposits in the western part of the Lufilian fold-and-thrust belt, Katanga (Democratic Republic of Congo). *J Afr Earth Sci* 46(5):455–469
- Duran CJ, Barnes SJ, Corkery JT (2015) Chalcophile and platinum-group element distribution in pyrites from the sulfide-rich pods of the Lac des Iles Pd deposits, Western Ontario, Canada: Implications for post-cumulus re-equilibration of the ore and the use of pyrite compositions in exploration. *J Geochem Explor* 158:223–242

- Duran CJ, Dubé-Loubert H, Pagé P, Barnes SJ, Roy M, Savard D et al (2019) Applications of trace element chemistry of pyrite and chalcopyrite in glacial sediments to mineral exploration targeting: example from the Churchill Province, northern Quebec, Canada. *J Geochem Explor* 196:105–130
- El Desouky HA, Muchez P, Boyce AJ, Schneider J, Cailteux JHL, Dewaele S, Quadt A (2010) Genesis of sediment-hosted stratiform copper–cobalt mineralization at Luiswishi and Kamoto, Katanga Copperbelt (Democratic Republic of Congo). *Miner Deposita* 45(8):735–763
- El Desouky HA, Muchez P, Cailteux J (2009) Two Cu–Co sulfide phases and contrasting fluid systems in the Katanga Copperbelt, Democratic Republic of Congo. *Ore Geol Rev* 36(4):315–332
- Esri (2018) OpenStreetMap contributors, and the GIS user community. Software: ArcGIS, ArcMap
- Faure G (1997) Principles and applications of geochemistry: Prentice Hall Upper Saddle River, NJ
- Foltyn K, Bertrandsson Erlandsson V, Zygo W, Melcher F, Pieczonka J (2022) New perspective on trace element (Re, Ge, Ag) hosts in the Cu–Ag Kupferschiefer deposit, Poland: insight from a LA-ICP-MS trace element study. *Ore Geol Rev* 143:104768
- Foster DA, Goscombe BD, Gray DR (2009) Rapid exhumation of deep crust in an obliquely convergent orogen: the Kaoko Belt of the Damara Orogen. *Tectonics* 28(4):1–24
- Franchini M, McFarlane C, Maydagán L, Reich M, Lentz DR, Meinert L, Bouhier V (2015) Trace metals in pyrite and marcasite from the Agua Rica porphyry-high sulfidation epithermal deposit, Catamarca, Argentina: Textural features and metal zoning at the porphyry to epithermal transition. *Ore Geol Rev* 66:366–387
- Frenzel M, Hirsch T, Gutzmer J (2016) Gallium, germanium, indium, and other trace and minor elements in sphalerite as a function of deposit type - A meta-analysis. *Ore Geol Rev* 76:52–78
- George LL, Cook NJ, Crowe BBP, Ciobanu CL (2018) Trace elements in hydrothermal chalcopyrite. *Mineral Mag* 2(1):59–88
- Glennie KW, Buller AT (1983) The Permian Weissliegend of NW Europe: the partial deformation of aeolian dune sands caused by the Zechstein transgression. *Sediment Geol* 35(1):43–81
- González-Jiménez JM, Yesares L, Piña R, Sáez R, de Almodóvar GR, Nieto F, Tenorio S (2022) Polymetallic nanoparticles in pyrite from massive and stockwork ores of VMS deposits of the Iberian Pyrite Belt. *Ore Geol Rev* 145:104875
- Gopon P, Douglas JO, Auger MA, Hansen L, Wade J, Cline JS et al (2019) A nanoscale investigation of Carlin-type gold deposits: An atom-scale elemental and isotopic perspective. *Econ Geol* 114(6):1123–1133
- Goscombe BD (1999) Geological report on magnetic and radiometric anomalies mapped in the region south of the Kunene Igneous Complex, Namibia. Company exploration reports. *Geol Surv Namibia*
- Goscombe BD, Gray DR, Hand M (2005) Extrusional Tectonics in the Core of a Transpressional Orogen; the Kaoko Belt. *Namibia J Petrol* 46(6):1203–1241
- Goscombe BD, Hand M, Gray DR (2003) Structure of the Kaoko Belt, Namibia: progressive evolution of a classic transpressional orogen. *J Struct Geol* 25(7):1049–1081
- Gregory DD, Large RR, Halpin JA, Baturina EL, Lyons TW, Wu S et al (2015) Trace element content of sedimentary pyrite in black shales. *Soc Econ Geol Spec Pub* 16:1389–1410
- Gregory DD, Cracknell MJ, Large RR, McGoldrick P, Kuhn S, Maslennikov VV et al (2019) Distinguishing ore deposit type and barren sedimentary pyrite using laser ablation-inductively coupled plasma-mass spectrometry trace element data and statistical analysis of large data sets. *Econ Geol* 114(4):771–786
- Guj P (1970) The Damara Mobile Belt in the south-western Kaokoveld South West Africa. PhD Thesis. University of Cape Town / Prec Res Unit, Cape Town, South Africa
- Hanson RE, Wardlaw MS, Wilson TJ, Mwale G (1993) U/Pb zircon ages from the Hook granite massif and Mwembeshi dislocation: constraints on Pan-African deformation, plutonism, and transcurrent shearing in central Zambia. *Prec Res* 63(3–4):189–209
- Heijlen W, Banks DA, Muchez P, Stensgard BM, Yardley BWD (2008) The nature of mineralizing fluids of the Kipushi Zn–Cu deposit, Katanga, Democratic Republic of Congo: Quantitative fluid inclusion analysis using laser ablation ICP-MS and bulk crush-leach methods. *Econ Geol* 103(7):1459–1482
- Hitzman MW, Broughton D, Selley D, Woodhead J, Wood D, Bull S (2012) The Central African Copperbelt: diverse stratigraphic, structural, and temporal settings in the world's largest sedimentary copper district. *Soc of Econ Geol Spec Publ* 16487–514
- Hitzman MW, Kirkham R, Broughton D, Thorson J, Selley D (2005) The sediment-hosted stratiform copper ore system. In *One Hundredth Anniversary Volume*: 609–642
- Hoffman PF, Halverson GP (2008) Otavi Group of the western Northern Platform, the Eastern Kaoko Zone and the western Northern Margin Zone. R. McG. Miller: The Geology of Namibia: Neoproterozoic to Lower Palaeozoic. 2nd ed. Windhoek, Namibia: *Geol Surv* 13:69–136
- Hu Y, Wei C, Ye L, Huang Z, Danyushevsky Leonid WH (2021) LA-ICP-MS sphalerite and galena trace element chemistry and mineralization-style fingerprinting for carbonate-hosted Pb–Zn deposits: Perspective from Early Devonian Huodehong deposit in Yunnan South China. *Ore Geol Rev* 136:104253
- Hulsbosch N, Hertogen J, Dewaele S, André L, Muchez P (2014) Alkali metal and rare earth element evolution of rock-forming minerals from the Gatumba area pegmatites (Rwanda): quantitative assessment of crystal-melt fractionation in the regional zonation of pegmatite groups. *Geochim Cosmochim Acta* 132:349–374
- John T, Schenk V, Mezger K, Tembo F (2004) Timing and PT evolution of whiteschist metamorphism in the Lufilian Arc–Zambezi Belt orogen (Zambia): implications for the assembly of Gondwana. *J Geol* 112(1):71–90
- Jowett EC (1987) Formation of sulfide–calcite veinlets in the Kupferschiefer Cu–Ag deposits in Poland by natural hydrofracturing during basin subsidence. *J Geol* 95(4):513–526
- Jowett EC (1986) Genesis of Kupferschiefer Cu–Ag deposits by convective flow of Rotliegendes brines during Triassic rifting. *Econ Geol* 81(8):1823–1837
- Jowett EC, Pearce GW, Rydzewski A (1987) A Mid-Triassic paleomagnetic age of the Kupferschiefer mineralization in Poland, based on a revised apparent polar wander path for Europe and Russia. *J Geophys Res: Solid Earth* 92(B1):581–598
- Kampunzu AB, Cailteux JHL, af Kamona, Intiomale MM, Melcher F, (2009) Sediment-hosted Zn–Pb–Cu deposits in the Central African Copperbelt. *Ore Geol Rev* 35(3–4):263–297
- Karnkowski PH (1999) Origin and evolution of the Polish Rotliegend Basin. *Polish Geol Inst Spec Papers* 3:1–93
- Kelvin M, Whiteman E, Petrus J, Leybourne M, Nkuna V (2022) Application of LA-ICP-MS to process mineralogy: gallium and germanium recovery at Kipushi copper–zinc deposit. *Minerals Eng* 176:107322
- Kiersnowski H, Buniak A (2006) Evolution of the Rotliegend Basin of northwestern Poland. *Geol Quart* 50(1):119–138
- Kłapciński J, Peryt TM (2007) Budowa geologiczna monokliny przedsudeckiej. *Monografia KGHM Polska Miedź SA*, 69–77
- Kłapciński J (1971) Lithology, fauna, stratigraphy, and paleogeography of the Permian in the Fore-Sudetic monocline. *Geol Sudet* 5:77–135
- Large RR, Halpin JA, Danyushevsky LV, Maslennikov VV, Bull SW, Long JA et al (2014) Trace element content of sedimentary pyrite as a new proxy for deep-time ocean–atmosphere evolution. *Earth Planet Sci Let* 389:209–220

- Large RR, Maslennikov VV, Robert F, Danyushevsky LV, Chang Z (2007) Multistage sedimentary and metamorphic origin of pyrite and gold in the giant Sukhoi Log deposit, Lena gold province Russia. *Econ Geol* 102(7):1233–1267
- Large RR, Danyushevsky L, Hollit C, Maslennikov V, Meffre S, Gilbert S et al (2009) Gold and trace element zonation in pyrite using a laser imaging technique: implications for the timing of gold in orogenic and Carlin-style sediment-hosted deposits. *Econ Geol* 104(5):635–668
- Liu Q, Mambwe P, Littke R, Muchez Ph (2024) Diagenesis and mineralization in the Central African Copperbelt, implications from the reflectance of pyrobitumen and Kübler (illite crystallinity) index. *Int J Coal Geol* 289:104534
- Liu S, Zhang Y, Ai G, Xue X, Li H, Shah SA et al (2022) LA-ICP-MS trace element geochemistry of sphalerite: metallogenic constraints on the Qingshitang Pb–Zn deposit in the Qinhang Ore Belt South China. *Ore Geol Rev* 141:104659
- Loftus-Hills G, Solomon M (1967) Cobalt, nickel and selenium in sulphides as indicators of ore genesis. *Miner Deposita* 2(3):228–242
- Magnall JM, Gleeson SA, Hayward N, Oelze M (2022) Using whole rock and in situ pyrite chemistry to evaluate authigenic and hydrothermal controls on trace element variability in a Zn mineralized Proterozoic subbasin. *Geochim Cosmochim Acta* 318:366–387
- Mambwe P, Swennen R, Cailteux J, Mumba C, Dewaele S, Muchez P (2023) Review of the origin of breccias and their resource potential in the Central Africa Copperbelt. *Ore Geol Rev* 156:105389
- Marsh EE, Hitzman MW, Leach David L (2016) Critical elements in sediment-hosted deposits (clastic-dominated Zn–Pb–Ag, Mississippi Valley-type Zn–Pb, sedimentary rock-hosted Stratiform Cu, and carbonate-hosted Polymetallic Deposits): a review. In *Rare Earth and Critical Elements in Ore Deposits* 18:307–322
- Master S, Rainaud C, Armstrong RA, Phillips D, Robb LJ (2005) Provenance ages of the Neoproterozoic Katanga Supergroup (Central African Copperbelt), with implications for basin evolution. *J Afr Earth Sci* 42(1–5):41–60
- McCann T, Pascal C, Timmerman MJ, Krzywiec P, López-Gómez J, Wetzel L et al (2006) Post-Variscan (end Carboniferous–Early Permian) basin evolution in western and central Europe. *Geol Soc Lon, Memoirs* 32(1):355–388
- McGowan RR, Roberts S, Boyce AJ (2006) Origin of the Nchanga copper–cobalt deposits of the Zambian Copperbelt. *Miner Deposita* 40(6–7):617
- McGowan RR, Roberts S, Foster RP, Boyce AJ, Coller D (2003) Origin of the copper–cobalt deposits of the Zambian Copperbelt: An epigenetic view from Nchanga. *Geology* 31(6):497–500
- Mederski S, Pršek J, Majzlan J, Kiefer S, Dimitrova D, Milovský R et al (2022) Geochemistry and textural evolution of As–Tl–Sb–Hg-rich pyrite from a sediment-hosted As–Sb–Tl–Pb±Hg±Au mineralization in Janjevo. *Kosovo Ore Geol Rev* 151:105221
- Melcher F, Erlandsson Bertrandsson V, Gartner V, Henjes-Kunst E, Raith J et al (2023) Carbonate-hosted “Alpine-type” Zn–Pb deposits in the Eastern and Southern Alps—trace element geochemistry and isotopic data of sulphides. *Irish-Type Deposits Around the world*. IAEG, Dublin, pp 443–478
- Mendelsohn F (1961) The geology of the northern Rhodesian copperbelt: Macdonald. London (UK), 523p
- Michalik M, Sawłowicz Z (2001) Multi-stage and long-term origin of the Kupferschiefer copper deposits in Poland. *Mineral Deposits at the Beginning of the 21st Century*: CRC Press, 235–238
- Mikulski SZ, Stein HJ (2012) The age of molybdenites in Poland in the light of Re–Os isotopic studies. *Biul - Państw Inst Geol* 452:199–216
- Mikulski SZ, Williams IS, Markowiak M (2019) Carboniferous–Permian magmatism and Mo–Cu (W) mineralization in the contact zone between the Małopolska and Upper Silesia Blocks (south Poland): an echo of the Baltica–Gondwana collision. *Int J Earth Sci* 108:1467–1492
- Miller RMG (2008) The geology of Namibia: Neoproterozoic to Lower Palaeozoic. 2nd ed. Windhoek, Namibia: Geol Surv 12–1:15–30
- Molak B (1995) Some structural and petrological aspects of the Cu (Co) mineralization in the Copperbelt and northwestern provinces of Zambia. *Annales des Sciences Géologiques: Royal Museum of Central Africa. Tervuren Belgium Sci Geol* 8:95–102
- Mukherjee I, Large R (2017) Application of pyrite trace element chemistry to exploration for SEDEX style Zn–Pb deposits: McArthur Basin, Northern Territory, Australia. *Ore Geol Rev* 81:1249–1270
- Muntean JL, Cline JS, Simon AC, Longo AA (2011) Magmatic–hydrothermal origin of Nevada’s Carlin-type gold deposits. *Nature Geosci* 4(2):122–127
- Onuk P, Melcher F, Mertz-Kraus R, Gäbler HE, Goldmann S (2017) Development of a Matrix-Matched Sphalerite Reference Material (MUL–ZnS-1) for Calibration of In Situ Trace Element Measurements by Laser Ablation–Inductively Coupled Plasma–Mass Spectrometry. *Geost Geoanal Res* 41(2):263–272
- Oszczepalski S (1999) Origin of the Kupferschiefer polymetallic mineralization in Poland. *Miner Deposita* 34(5):599–613
- Oszczepalski S (1989) Kupferschiefer in southwestern Poland: sedimentary environments, metal zoning, and ore controls. *GAC Spec Pub* 36:571–600
- Paton C, Hellstrom J, Paul B, Woodhead J, Hergt J (2011) Iolite: Free-ware for the visualisation and processing of mass spectrometric data. *J Anal at Spectrom* 26(12):2508–2518
- Paul J (2006) Der Kupferschiefer: Lithologie, Stratigraphie, Fazies und Metallogenese eines Schwarzschiefers. *Z Dt Ges Geowiss* 157(1):57–76
- Peryt TM, Durakiewicz T, Kotarba MJ, Oszczepalski S, Peryt D (2012) Carbon isotope stratigraphy of the basal Zechstein (Lopingian) strata in Northern Poland and its global correlation. *Geol Quart* 56(2):285–298
- PGI, Polish Geological Institute (2024) The balance of mineral resources deposits in Poland (in Polish). Bilans zasobów złóż kopalin w Polsce. Stan na 31 XII. 2023 ISSN 2299–4459
- Podemski M (2001) Palaeozoic porphyry molybdenum–tungsten deposit in the Myszków area, Southern Poland. *Polish Geol Inst Spec Papers* 6:1–88
- Pohl WL, Biryabarema M, Lehmann B (2013) Early Neoproterozoic rare metal (Sn, Ta, W) and gold metallogeny of the Central Africa Region: a review. *Appl Earth Sci* 122(2):66–82
- Porada H (1989) Pan-African rifting and orogenesis in southern to equatorial Africa and eastern Brazil. *Prec Res* 44(2):103–136
- Rainaud C, Master S, Armstrong RA, Phillips D, Robb LJ (2005) Monazite U–Pb dating and ⁴⁰Ar–³⁹Ar thermochronology of metamorphic events in the Central African Copperbelt during the Pan-African Lufilian Orogeny. *J Afr Earth Sci* 42(1–5):183–199
- Richards JP, Krogh TE, Spooner ETC (1988) Fluid inclusion characteristics and U–Pb rutile age of late hydrothermal alteration and veining at the Musoshi stratiform copper deposit, Central African copper belt Zaire. *Econ Geol* 83(1):118–139
- Sangster DF (2002) The role of dense brines in the formation of vent-distal sedimentary-exhalative (SEDEX) lead–zinc deposits: field and laboratory evidence. *Miner Deposita* 37(2):149–157
- Schneider J, Melcher F, Brauns M (2007) Concordant ages for the giant Kipushi base metal deposit (DR Congo) from direct Rb–Sr and Re–Os dating of sulfides. *Miner Deposita* 42:791–797

- Schwartz MO (2000) Cadmium in zinc deposits: Econ Geol of a polluting element. *Int Geol Rev* 42(5):445–469
- Selley D, Broughton D, Scott RJ, Hitzman M, Bull SW, Large RR et al (2005) A new look at the geology of the Zambian Copperbelt. *Soc Econ Geol*, 100. *Economic Geology 100th Anniversary Volume*: 965–1000
- Speczik S, Puttmann W (1987) Origin of Kupferschiefer mineralization as suggested by coal petrology and organic geochemical studies. *Acta Geol Pol* 37(3–4):167–188
- Strengel-Martinez M, Speczik S, Jankowski J (1993) Epigenetic veins in Kupferschiefer deposits of SW Poland. *Arch Mineral* 49(1):203–204
- Speczik S, Zieliński K, Bieńko T, Pietrzela A (2021) The prospecting strategy for a deep Cu–Ag ore deposit in Poland—an anatomy of success. *Ore Geol Rev* 131:104053
- Swardt AM (1962) Structural relationships in the N. Rhodesian Copperbelt: an alternative explanation. *Northern Rhodesia Geol Surv Occas Paper* 30:15–28
- Sykora S, Cooke DR, Meffre S, Stephanov AS, Gardner K, Scott R et al (2018) Evolution of pyrite trace element compositions from porphyry-style and epithermal conditions at the Lihir gold deposit: Implications for ore genesis and mineral processing. *Econ Geol* 113(1):193–208
- Symons DTA, Kawasaki K, Walther S, Borg G (2011) Paleomagnetism of the Cu–Zn–Pb-bearing Kupferschiefer black shale (Upper Permian) at Sangerhausen. Germany *Miner Deposita* 46(2):137–152
- Taylor CD, Causey JD, Denning P, Hammarstrom JM, Hayes TS, Horton JD et al (2013) Descriptive models, grade-tonnage relations, and databases for the assessment of sediment-hosted copper deposits: with emphasis on deposits in the Central Africa Copperbelt, Democratic Republic of the Congo and Zambia: Chapter J in *Global mineral resource assessment*. US Geol Surv Sci Invest Rpt 2010–5090–J, 154p
- Tomaszewski JB (1978) The geological structure of the Lubin-Sieroszowice region (Lower Silesia). *Geol Sudetica* 13:85–132
- Torró L, Millán-Núñez AJ, Benites D et al (2023) Germanium- and gallium-rich sphalerite in Mississippi Valley-type deposits: the San Vicente district and the Shalipayco deposit, Peru. *Miner Deposita* 58:853–880
- Torró L, Benites D, Vallance J, Laurent O, Ortiz-Benavente BA, Chelle-Michou C, Proenza JA, Fontboté L (2022) Trace element geochemistry of sphalerite and chalcopyrite in arc-hosted VMS deposits. *J Geochem Explor* 232:106882
- Unrug R (1988) Mineralization controls and source of metals in the Lufilian fold belt, Shaba (Zaire), Zambia, and Angola. *Econ Geol* 83(6):1247–1258
- USGS (2020) Mineral commodity summaries 2020. U.S. Geological Survey, Reston, VA: 204p
- Van Wilderode J, Debruyne D, Torremans K, Elburg MA, Vanhaecke F, Muchez P (2015) Metal sources for the Nkana and Konkola stratiform Cu–Co deposits (Zambian Copperbelt): insights from Sr and Nd isotope ratios. *Ore Geol Rev* 97:127–138
- Van Wilderode J, El Desouky HA, Elburg MA, Vanhaecke F, Muchez P (2014) Metal sources for the Katanga Copperbelt deposits (DRC): insights from Sr and Nd isotope ratios. *Geol Belg* 17(2):137–147
- Van Wilderode J, Heijlen W, Muynck D, Schneider J, Vanhaecke F, Muchez P (2013) The Kipushi Cu–Zn deposit (DR Congo) and its host rocks: A petrographical, stable isotope (O, C) and radiogenic isotope (Sr, Nd) study. *J Afr Earth Sci* 79:143–156
- Vaughan DJ, Sweeney MA, Friedrich G, Diedel R, Haranczyk C (1989) The Kupferschiefer; an overview with an appraisal of the different types of mineralization. *Econ Geol* 84(5):1003–1027
- Velojić M, Bertrandsson Erlandsson V, Melcher F, Onuk P, Jelenković R, Cvetković V (2022) Trace elements in pyrite from the Čukaru Peki porphyry Cu–high-sulfidation deposit, Serbia: implications for ore evolution in a polyphase hydrothermal system. *Geol Croatica* 75:303–316
- Waele B, Johnson SP, Pisarevsky SA (2008) Palaeoproterozoic to Neoproterozoic growth and evolution of the eastern Congo Craton: its role in the Rodinia puzzle. *Prec Res* 160(1):127–141
- Wedepohl KH, Delevaux MH, Doe BR (1978) The potential source of lead in the Permian Kupferschiefer bed of Europe and some selected Paleozoic mineral deposits in the Federal Republic of Germany. *Contrib Mineral Petrol* 65(3):273–281
- Wilson SA, Ridley WI, Koenig AE (2002) Development of sulfide calibration standards for the laser ablation inductively-coupled plasma mass spectrometry technique. *J Anal at Spectrom* 17(4):406–409
- Wodzicki A, Piestrzyński A (1994) An ore genetic model for the Lubin—Sieroszowice mining district Poland. *Miner Deposita* 29(1):30–43
- Xie Z, Gopon P, Xia Y et al (2023) Does SW China have Carlin-type gold deposits? A micro- to atomic-scale perspective. *Miner Deposita* 59(4):757–772

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